

Title: Site Procedure for Permit to Work (PTW)

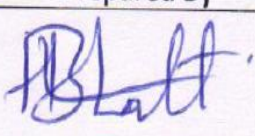
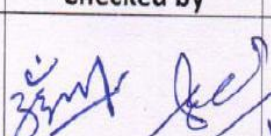
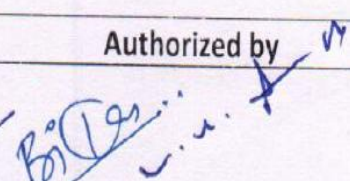
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Distribution: GFL Dahej plant HOD, Control room and & Sustainability internal portal

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ABBREVIATIONS

BA	Breathing apparatus
CES	Centre engineering service
COR	Corporate
EHS	Environment, Health, Safety
EHSS	Environment, Health, Safety & Sustainability
ERP	Emergency response procedure/Plan
GFRC	Gujarat Fluorochemical Research Centre
HOD	Head of Department
HR	Human Resource
HSE	Health Safety Environment
IMS	Integrated management System
IS	Indian Standards
JSA	Job safety Analysis
LEL	Lower explosive limit
LOTO	Lock-out Tag-out
MOC	Management of Change
NA	Not Applicable
NDO	Night Duty Officer
OHC	Occupational Health and Safety
P-JSA	Pre-Job safety Analysis
PM	Preventive maintenance
PPE	Personal Protective Equipment
PTW	Permit to Work
Rev	Revision
SD	Safety District
SMP	Standard Maintenance Procedure
SOP	Standard Operating Procedure
TLD	Thermo-luminescent dosimeter
SRP	Standard Rules and procedure
AGL	Act Group Leader
SDS	Safety data Sheet
SDO	Special Duty officer

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DEFINITIONS

These definitions are provided to describe the intend for this procedure. Meaning /definitions of these terms might be different for other purposes.

GFL Complex:

Means all the operating areas of various Plants and units of Fillings, Power, Utilities, Loading and unloading areas, Off-site pipelines, pipe racks and crossings, corridor, tank farms and other associated service areas and buildings such as Laboratory, Administrative Building, Stores etc. at Dahej.

Safety District:

It is predefined demarked and approved area. This has been approved by the unit head as per document number GFL-PP-DWG-SD-04-15-D0127.

In the some of the safety districts small area/equipment are being operated/maintain by other HOD. In this case the particular area will be under ownership of the operating HOD of the equipment.

Routine activities:

The Activities which are regular operations and are controlled through established procedures do not require permit. These activities should to be well defined by each department with proper SOP / SMP/ checklist (Document number must in IMS document).

Non-functional Area:

Means Engineering Stores, canteen building, HR Building, Material gate, OHC, security gate, weight bridge station, Offices, GFRC and laboratories.

Non-Routine non-functional activities:

The activities other than routine which may or may not have well established procedures. e.g. Change management (MOC).

Permit/PTW/Permit-to-Work:

Means a written certificate in prescribed format allowing certain work to be carried out in a fixed place and during a given period of time, or a certificate allowing entrance and stay in certain facilities.

Cold work:

All jobs that do not fall under purview of a hot work permit require a cold work permit.

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Hot work:

Any work which involves welding, burning, soldering, brazing, blasting, gas cutting, chipping by spark production tools, use of certain power-driven tools, non-flame-proof electrical equipment or equipment with internal combustion engines and including any other work which is likely to produce sufficient heat capable of igniting flammable gases.

Some examples of hot work are:

- Welding or flame cutting equipment.
- Grinder, electric drill or other non-flameproof electrical equipment
- Hammering, chipping, drilling (particularly in the presence of flammable/ combustible/ explosive atmospheres)
- Spark ignition engines or non-approved compression ignition engines
- Hand tools that may generate a spark.
- Any device, which is likely to produce a source of ignition or be an ignition source, e.g., cell phones, pagers, radios, cameras/camcorders etc.

Work at Height:

Work at height (permit is required) to perform any (maintenance / inspection) work above/below 1.8 mtr elevation from ground level where permanent platforms / railings are not available.

Some examples "working at height:

- Working on trestles
- Working on a (flat) roof
- Erecting falsework (Falsework refers to temporary structures used in construction to support a permanent structure) or formwork (Formwork refers to temporary molds or structures used to shape and support concrete until)
- Working on a ladder
- Working at ground level adjacent to an excavation.
- Working on formwork within an excavation
- Working near or adjacent to fragile materials

Confined Space Entry:

A Confined Space is any enclosed or partially enclosed space, either above or below ground, where there is some risk of reduced oxygen level, or accumulation of toxic flammability or explosive materials, or entry or exit are limited or there is a risk of engulfment.

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Excavation Permit:

Means digging, road cutting, removing soil, chipping, grading, any excavation work which involves excavation of ground 1 Ft or more deep either using mechanical equipment or manually, site preparation.

Line Breaking:

Intentional opening of a pipe, line, or duct that contains or has contained material capable of causing injury It requires that all members of a line breaking team understand the hazards related to the material and equipment involved.

Hydraulic & Pneumatic System Work:

Any machine & equipment where hydraulic energy, like hydraulic jaw, hydraulic doors, chambers, cutters, hammers, pusher and lifting hydraulic tools where pneumatic energy is stored & used for operating, functioning of the machine, Many hydraulic & pneumatic operated equipment working require a full hazard analysis during maintenance & repair as store hydraulic/pneumatic energy of the equipment having high potential hazard for accident if release suddenly, so release the hydraulic pressure or pneumatic pressure of related in an operating system prior to removing or adjusting components. The hot fluid in the system is under extreme pressure. If a worker is exposed to hydraulic fluid while removing or adjusting components, he or she may sustain burns, bruises, cuts, abrasions, or injection of fluid into the skin. Do not rely solely on the hydraulic lift if you must work on hydraulic components with the system raised. Set the working unit on blocks as a precautionary measure.

Radiography:

Industrial radiography uses radiation to assess the integrity of structure. Industrial Radiography Permits are issued to each service provider for a defined period of time as needed by the service provider.

Electrical Isolation:

Means positive isolation of equipment connected to an electrical power system for the purpose of maintenance and installation.

Blanket Permits:

A single PTW given for multiple jobs at different location or same job at multiple location. Blanket permit strictly prohibited. For e.g 1) A single permit is given for PM of Pump A at X location and Pump B at Y location.

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Permit issuer:

A Permit Issuer is a person who is engineer/ shift In-charge, has experience of total one year plus 3-month experience in same plant. He has got trained, (PTW standard, JSA Standard, & ERP), competent, and authorized to issue a Permit to Work as per permit matrix Annexure format. Has gone through MOC(P).

Note: matrix approved by Plant HOD.

Permit Acceptor:

A person who is an engineer accepting the permit. Person who is responsible to handle work. He/She shall have experience of total 1 year plus 3 months in same plant. Is trained (PTW standard, JSA Standard, & ERP), competent, and authorized to accept a Permit to Work as per permit matrix. Annexure format. Has gone through MOC(P).

Note: PTW matrix approved by Plant HOD.

Permit Approver-I:

A person who is L4 (Manager) and above level, can approve the permit as Approver-I. He/ She shall have minimum experience of total 5 years including 3 months in the same area/plant. He/ She has to complete the self-study of training module of PTW, JSA, CSE, Hot Work and ERP. Has gone through MOCP Process. He can approve all the permits of plant/area/equipment which are under his jurisdiction.

Note: PTW matrix approved by Dy Unit Head/Unit Head.

Permit Approver-II:

A person who is L3 (Plant HOD level) can approve the permit as Approver-II. He/ She shall have experience of 3 months in same plant/area. He/ She has to complete the self-study of training module of PTW, JSA, CSE, Hot Work and ERP. He should have gone through MOCP process. He can approve all the permit of plant/area/equipment which are under his jurisdiction.

On w/off, Holiday or non-availability of Approver I, He can approve all the permit of plant/area/equipment which are under his jurisdiction as Approver I & Approver II.

Note: PTW matrix approved by Dy Unit Head/Unit Head.

Permit Approver-III :

A person who is L2 (VP/AVP/Sr.VP) can approve the permit as Approver-III. He/ She shall have experience of 3 months in same plant/area. He/ She has to complete the self-study of training module of PTW, JSA, CSE, Hot Work and ERP. He should have gone through MOCP process.

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The responsibilities of this role are:

In case of confined space/Vessel entry permit extended to C shift or on non-working day, the permit shall be approved by Approver-III after informing Dy UH/Unit head.

On w/off, Holiday or non-availability L-3, He can approve all the permit of plant/area/equipment which are under his jurisdiction as approver II & Approver III.

NOTE: Jobs requiring Approver-III is discouraged. Hence such jobs to be avoided unless it is extremely urgent.

Permit Matrix:

A list of duly authorized persons who will perform the duties of issuer, acceptor, gas tester, and approvers (Matrix format as per annexure).

Fire Watch:

A fire watch is a person who will oversee any probability of fire due to hot surfaces or spillage of sparks during the hot work and for 30 minutes after job completion. He/ She could be the contractor supervision and may work as fire watch for more than one job in his vicinity. He/ She should have basic knowledge of operating fire extinguisher and able to communicate clearly.

Standby Person:

A Standby Person must be always available at the entry during confined space work. His name must be mention in the permit. He/ She should able to:

- ✓ Communicate clearly.
- ✓ Control the exit and entry of person and keeping the record.
- ✓ Raise a call for help in case of any distress call/ raised by persons working inside.
- ✓ Alert and call for evacuation the people inside in case of outside emergency.

Gas Tester:

A designated person who has received training for the operation of gas testing machines/instrument and procedure of gas testing.

He/ She is responsible for:

- ✓ Carrying out gas testing before the starting and during the jobs. Keep record of the gas testing data on PTW.
- ✓ A gas Tester carries out Gas test for oxygen content, LEL and toxic content.
- ✓ LEL concentration within 5.0 m radius of the work location for welding jobs/ hot work.

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Self-Permitting:

In a Permit to Work (PTW) system, self-permitting refers to the practice where an individual authorizes their own work permit without requiring approval from a separate, designated authority. (e.g Mr. XYZ is issuing the permit to Mr XYZ for Hot Work)

Isolation:

A physical barrier between a source of energy (mechanical, electrical, hydraulic or static head) or process materials.

Live Permit:

A permit for a work that is active and ongoing.

Method Statement:

A breakdown of an activity into individual tasks.

Monitoring:

Routine checking of permits and controls in the course of the work.

Revalidation:

Reassessing the work site and PTW controls to determine if work can continue safely for another set period of time.

Review:

Examining the fundamental design of the system to see whether it should be changed in the light of experience.

Shift Handover:

Process of transferring responsibility for controlling and monitoring work from the present Operations shift to the oncoming shift.

Supporting Certificate Signatory:

The person responsible for confirming that the hazards have been (Validated, Authorized Safety correctly identified and assessed and, that the proposed controls area inspector, Fire Permit Signatory) adequate.

Suspension:

Temporary invalidation of a permit for a period during which the work may not proceed.

Validity Period:

The period during which a permit may remain active, and after which the work site controls must be reassessed.

Worker (User):

A member of the work party, working to the instruction of the Permit Holder.

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Gas Test - For Hot Work:

A signed statement on PTW by an authorized Gas Tester experienced in gas monitoring, that a gas test has been taken for the presence of flammable vapors/toxic gases within a specified location and that the concentration shows a nil LEL response.

Gas Test for Confined Space:

A signed statement by an Authorized Gas Tester for/toxic gases, flammable gases and oxygen tests have been taken within the confined space to be entered. Prior to entering a confined space (without the assistance of long line breathing apparatus) the toxic gas vapors shall show nil, LEL- below 5% and oxygen concentration shall be between 19.5 and 23.5 % by volume. The frequency of testing shall be specified and the certificate re-signed after each test.

Hazard: -

The potential a substance, activity, event, process or energy source has, to impair the health or well-being of a person or the environment.

Risk: -

The probability that a hazard may fulfill its potential.

Lower Explosive Limit (LEL):-

The concentration of flammable gas, vapor or mist in air below which an explosive gas atmosphere will not be formed.

1. PURPOSE:

The purpose of this procedure is to provide clarity and helps in

- Aligning the requirement for PTW process with Corporate Standard and other Statutory Requirement.
- To provide a guideline for completing permit to work (PTW) perfectly.
- To provide guideline for uniform understanding about PTW & its compliance.

2. SCOPE:

This Procedure is applicable to all non-routine jobs that are carried out in the premises of M/s Gujarat Fluorochemicals Limited at Dahej site.

3. PROCEDURE:

3.1 Preparation of Permit Matrix and training records:

Every Plant HOD has to prepare the Permit Matrix and keep training records (both Format in annexure) of all eligible Permit Issuers, Acceptors, Approvers, Gas Testers. Eligibility criteria for all these functions is elaborated in respective sections. Keep it readily available (preferably displayed in the control room/field room).

3.2 Identification of jobs which require PTW

Every Plant has to prepare a list of all routine and non-routine activities. Clearly specify which activity require a permit and which do not. Ensure that the activities list is available in the control room in both soft or hard copy formats. (Activity list to be prepare as per format given in annexure- Routine

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and non-routine activities) Jobs, routinely carried out following an approved SOP, do not require the permit and non-routine jobs required the permit.

Note:

Generally, routine jobs do not require permit, however, few high-risk potential jobs (like tanker loading /unloading require a PTW. Respective HOD has to identify such jobs which require a permit.

3.3 Listing out the Specific PPEs Requirement.

List PPE Required for handling the chemicals used in the plant. Ensure that the necessary PPE is readily available whenever needed.

Every Plant has to prepare a list of specific PPEs required for their plant based on the activities carried out and chemicals being handled.

3.4 Job planning and Identification of Jobs to be carried out

A job planning meeting is to be carried out every day evening in every plant. Operations and maintenance team has to decide the jobs for the next day, jointly. Few of the criterion for the job selection, for the next day, could be urgency of the job- impact on the safety, availability of manpower, spares etc.

Note: Emergency Jobs can be started without Permit and can be regularized after emergency handling. (Emergency e.g Chlorine leakage, Fire, chemical leakage etc....)

3.5 Preparation of JSA

Team of permit issuer, acceptor & Contractor Supervisor jointly visit the job site and prepare the JSA (Format No HSE/SAF/F/05). Permit Issuer & Acceptor to write the JSA.

- **Sequence of basic job steps - list-down the activities/sub-activities.**
- **Potential Hazard** - Identify potential hazards associated with all listed activities.
- **Risk** - Identity the risk associated with all the activities.
- **Safeguard/Control** - Suggest the control measure to mitigate/Control potential hazards.
- **Assign responsibility** for each suggested control measure.
- The JSA team should sign the document upon completion.

Note:

1. Refer JSA site procedure & **CORPORATE HEALTH & SAFETY STANDARD - GFL/COR/EHSS 2002.00-2.**
2. **Hazard identified in JSA and suggested control measures should align with the PTW section E.**

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3.6 Completion of PTW Format

Section wise details are given below:

Section A: Type of Work Permit

Tick the appropriate type of work permit based on nature of job.

Section B: Job Details

Specify - Plant/Unit, Area/location, SAP Notification Number, Equipment Name with tag number, Job description. Provide details of Job execution agency, Executing department and number of person/s at work.

Section C: Nature of Work

Specify the nature of Work.

Section D: Tool and Equipment to be used

Specify the tools/equipment required to carry out the work.

Section E: Hazard Consideration

Specify hazards considered in JSA sheet for the job.

Section F: Job and Equipment Preparation.

Specify the compliance ensured to execute the job.

Section G: PPEs and Fire Protection

Specify the job specific PPE's required. (Refer SDS for selection of PPEs for chemicals being involved with the job). Specify appropriate fire protection if required.

Section H: Isolation.

1) Electrical Isolation

Refer electrical LOTO procedure.

Note: Electrical isolation key must be kept with executing agency.

2) Other Isolation

Process Isolation: Ensure the system is free from hazardous material, pressure etc.

- Depressurized the system by suitably draining/venting of process material in a safe manner.
- Wash/de-contaminate the system.
- Positive isolation (blinding), locking & tagging

Mechanical Isolation

- Locking of rotating parts by clamping/chain etc.
- Draining of other energy sources such as depressurization, hydraulic, kinetic energy prior to locking.
- Ensure the mechanical isolation is done (For positive isolation standard tail blind to be used/Double block & bleed. OR remove the connected line) and the permit is signed by the person isolated the system.

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Section I: GAS Test: Accepted value of gas testing result.

Allowable limit for gases		
Oxygen	Toxic Gases	Flammable Gases
19.5% to 23.5 %	Zero	Zero

General Guidelines for Gas Tester:

- Authorized Gas Tester:** Gas testing is to be done by an authorized gas tester (trained, approved and listed in the Permit Matrix).
- Equipment Checking:** Ensure that the gas detection equipment/instrument is calibrated and suitable for detecting the specific gases that may be present. It is suggested to read OEM/manual/ brochure for gas testing equipment.
- Personal Protective Equipment (PPE):** Wear appropriate PPE, such as gloves, safety glasses, and a gas mask if necessary.
- Area Inspection:** Inspect the area where the hot work will be performed to identify potential sources of gas.
- Gas Detector Setup:** Turn on the gas detector as per manufacturer's instructions.
- Baseline Reading:** Take a baseline reading in a safe area to ensure the detector is functioning correctly.
- Test at Different Levels:** Conduct the gas test at various levels (e.g., ground level, mid-level, and high level) to detect any stratified gases.
- Wherever forced ventilation is used, gas test should be carried out after minimum 10 min of stopping the educator/fan/blower.
- Gas testing in confined spaces must be revalidated before reentry if there has been a break of more than one hour.
- If there is any abnormality found the work must be stopped immediately and inform the permit issuer and responsible person.

Frequency of Gas testing:

For Hot Work:

Gas testing shall be carried out at least once every 2 hours within 15 M of the work site by the authorized gas tester if hot work is at close proximity of area where there is potential fire and/or explosion risk.

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For Confined Space Entry:

1. Entry to the confined to made **within 1 hrs. of gas testing** otherwise conduct gas testing again.
2. A periodic gas testing, using portable gas testing instrument to be done, **every 2 hrs.**
3. **It is suggested** If Continuous gas monitor is available is to be used inside the confined space to quickly detect the change in atmospheric conditions inside the Confined Space.

Oxygen concentration:

- Wherever forced ventilation is used. The gas test should be carried out after minimum 10 min of stopping the educator/fan/blower.
- Gas testing in confined spaces must be revalidated before reentry if there has been a break of more than one hour.
- If there is any abnormality found the work must be stopped immediately and inform the permit issuer and responsible person.

NOTE:

- Carry out regular gas test of confined spaces, where there is possibility of gas generation due to nature of work, being performed inside, such as removal of sludge, hot works, etc.

Section J, K, L & M

Following sequence must be followed for getting signature & job clearance on permit.

Sequence of approval for issuing a permit: (1 means 1st signing person 2 means 2nd signing person and so on.)

Type of Permit	Cold Work	Hot Work	Confined Space	Excavation	Rigging job	Work at Height	Line Breaking	Radiography	Hydraulic/Pneumatic System
Section: J Permit Issuer	1	1	1	1	1	1	1	1	1
Section: H Isolation	2	2	2	2 Excavation Clearance (Section F)	NA	NA	2	2	NA
Section: M									
Approver I	-	3	3	3	3	2	3	3	-
Approver II	-	4	4	4	4	-	4*	-	-
Section: L Permit Acceptor	3	5	5	5	2	3	5	4	2
Section: K Area operator	4	6	6	6	5	4	6	5	3

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*** Defined chemicals with hazards**

Section N

Following sequence must be followed for permit closing/ cancellation.

Sequence of permit closing: (1 means 1st signing person 2 means 2nd signing person and so on.)

Type of Permit	Cold Work	Hot Work	Confined Space	Excavation	Rigging job	Work at Height	Line Breaking	Radiography	Hydraulic/ Pneumatic System
Permit Acceptor	1	1	1	1	1	1	1	1	1
Area Operator	2	2	2	2	2	2	2	2	2
Permit Issuer	3	3	3	3	3	3	3	3	3
Power Restored	4	4	4	4	NA	NA	4	4	NA

- When the work is complete, all copies of the permit should be reunited at the issuing point.
- The Permit Issuer or his delegate and the Permit Holder should then inspect the work site and jointly confirm that they are satisfied that the work has been completed as intended and the worksite has been left in a safe and tidy condition. Only then the Permit Issuer should sign off the permit.
- After completion of jobs executing department to give isolation key to permit issuer. After conformation all the required jobs are done permit issuer will call electrical department to energize the equipment.
- In case of any trial required after electrical energization permit issuer, permit acceptor and electrical person must be present and do trial checkup.

Note: Anyone can stop working on site if the job execution is violating agreed permit conditions. Reason to stop the work or closure must be written on permit.

3.7 Permits are not applicable for the following areas / activities:

- Resetting of electrical trips and alarms.
- Routine work like Tanker loading and unloading (connection / disconnection) and any other routine activities which are done by qualified trained persons as per SOP, very frequently.
- Truck loading unloading /Scrap loading unloading.
- Conditional monitoring of running equipment's by CES Department personnel as they have defined SOP.
- Identified and approved by Unit Head/Deputy Unit Head as non-permit area/job.

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3.8 Permit Validity, Extension and Approver:

- All permit to be revalidated during every shift change and has to sign on the permit.
- Permit to be revalidate if any issuer, operator or acceptor change during work and has to sign on permit.
- All permits issued by the A-shift engineer need to be extended before 15:45 hrs. B- Shift team has to sign the permit.
- All permits issued by the G shift engineer that need to be extended after 17:40 Hrs, must be revalidated and signed by the B shift engineer within one Hrs.
- If both the Head of Department (HOD) and the assigned L4 person are not available (during w/off /Holiday), PTW will be approved by SDO/NDO after obtaining approval from the concerned plant HOD via email.

3.8.1 PTW Validity and Extension

Type of Permit/Validity	Cold Work Permit & Height Work	Hot Work/Confine Space/ /Excavation/Rigging Job/Line Braking /Radiography/ Hydraulic /Pneumatic System.
VALIDITY	08 Hrs. from issue time or shift end whichever is earlier	08 Hrs. from issue time or shift end whichever is earlier
EXTENSION	7 days if the condition of work is remained unchanged (Review and signed JSA every day)	24 Hrs. for the same day but extend in each shift starting

Note: For Project work within the operating plant permit validity, Approval requirement & extension will remain the same.

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3.8.3 Permit Approval

	9:00 Hrs to 17:30 Hrs	17:30 Hrs to 23:00 Hrs	23:00 Hrs to 09:00 Hrs
Cold Work	Approver Sign Not Required	Approver Sign Not Required	Approver Sign Not Required
Hot Work	Approver I And Approver II	Approver I And Approver II	Approver II
Confined Space	Approver I And Approver II	Approver I And Approver II	Approver III
Excavation	Below 1.8 Approver I And Approver II	Below 1.8 Approver I And Approver II	Below 1.8 Approver II
Work at Height	Approver I	Approver I	Approver II
Line Breaking	Approver I And Approver II	Approver I And Approver II	Approver II
Radiography	Approver I	Approver I And Approver II	Approver II
Rigging job	Approver I	Approver I And Approver II	Approver II

Note:

- 1) 23:00 Hrs to 09:00 Hrs: (if fresh permit is being issued /extension is required then NDO may be requested to sign with communication on the mail by Approver II.
- 2) Approver I (L4) is authorized as Approver II in absence of Approver II (HOD). Approver I name should be included as Approver II in permit matrix.
- 3) Rigging Plan must be approved as per checklist HSE/SAF/F/27
 - i) For load ≤ 15 MT rigging plan should be approval by Maintenance Engineer, Maintenance Manager and Maintenance HOD.
 - ii) For load > 15 MT rigging plan should be approval by Maintenance Engineer, Maintenance Manager, Maintenance HOD, CES / Project head. For load > 30 MT additional Engineering Head approval required.

3.8.4 Permit Extension During Annual Shutdown:

Same as per regular Procedure, however permit relaxation in permit approver can be taken from unit head on mail.

3.9 PTW Rule for Hot Work at Contractor's Fabrication Yard Located Inside Plant.

3.9.1 Permit Issuance and Validity:

- Permit will be issued on Monday and will be valid from Monday to Saturday.

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- On Sunday, only essential job will be done with prior permission. Issuing and approval authorities will be as per safety districts.

3.9.2 Specific Locations and Responsibility:

For hot work activity at contractor's shed near CPP- South, CaCl₂- East, central work shop:

3.9.2.1 Permit issuer: Mechanical Engineer from Central Engineering.

3.9.2.2 Permit Approver: HOD of Central Engineering.

3.9.2.3 Permit Receiver: Contractor Supervisor/other Mechanical Engineer. (Note: He/ She must not be issuer)

3.10 PTW rule for operating equipment in other area.

At Dahej site, there are some Safety Districts in which a part of the area is being operated /used by other departments/district owners. In such cases, following clarification is to be used.

3.10.1 Issue of Permit to be done by user.

3.10.2 Maintenance and preventive maintenance to be done by user maintenance team.

For Example: -1) PFA MMT location in PTFE: Issuer PFA team & Receiver PFA Maintenance team.

1) TFE 1 Electrical Transformer: Issuer Electrical Engineer & receiver Maintenance electrical team.

2) AHU location in PTFE plant: Issuer Utility Engineer & Maintenance Utility Mechanical Team.

Note: All such permit must be counter sign by safety district owner/area issuer.

3.11 Road Closer & PTW: The Respective plant HOD has to take approval for road closer and alternate route from HSEF, Security, and Logistics teams via email at least 24 hours in advance.

3.12 Permit-To-Work in Electrical Substation/ Electric Sub-district/Electrical Transformer, Panels room.

- PTW is issue by department electrical engineer & accepted by the electrical engineer (executor).
- PTW is to be counter signed by process shift engineer of respective Safety District /affecting the plant.

Note: 1) Electrical department has to prepare permit matrix as per format given in annexure.

2) For further clarification, please refer Electrical Safety Standard and Site Procedure.

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3.13 PTW in Utility Area:

- PTW is issue by Department Utility Engineer & Accepted by the Utility Engineer (executor).
- PTW is to be counter signed by process shift engineer of respective Safety District /affecting the plant.

Note: 1) Utility department has to prepare permit matrix as per format given in annexure.

3.14 PTW Management for the Process Related Activities (where mech/elect/Inst department is not involved)

- PTW is issue by the Operations Engineer and accepted by other authorized Process Engineer.

3.15 Fire Water Line Related Work

Please follow HSEF guidelines.

3.16 The Work Permit Format distribution

- A minimum of two copies of the permit are needed:
- The Permit Acceptor copy displayed on site.
- The Permit Issuer copy held at the issuing point.
- It is important that the Permit holder of the permit has a copy readily available for reference by the work group and by persons conducting inspections of the work. The permit, and any supporting documents, should therefore be displayed at the work site.

4.0 PTW DETAILS FOR VARIOUS TYPES OF JOBS:

In case of Conflict between PTW & individual procedure, individual procedure to be consider final.

4.1 Work at Height:

Work at height permit is required to perform any Maintenance / inspection work at and above 1.8 meters elevation from ground level where permanent platforms / railings are not available.

Examples of “Working at Height” are:

1. Flat roofs, slopping roofs, fragile roof
2. Structure erection/ special structure
3. Working on pipe racks and cable trays,
4. Working on flat and sloping tank roofs,
5. Working on tall structures like lighting towers, chimneys, electrostatic precipitators, transmission towers, cooling towers etc.

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6. Insulation of piping at height
7. Confined space work at Height
8. Loading and unloading of materials from trucks
9. Working near Excavations
10. Working on Bucket Trucks and Man lifts
11. Painting works inside or outside the building

The Work at height permit issuer is requires to ensure that:

- A P-JSA is carried out for all work conducted at height.
- People working at a height are medical fit.
- Equipment used for work at height is properly inspected and maintained.
- Risks from fragile surfaces are properly controlled.
- The P-JSA should include a careful examination of what harm could be caused from working at height with a view to taking the effective steps to reduce the likelihood of this harm occurring, either through avoiding the activity or, where this is not reasonably practicable, by carrying it out in a safe manner using work equipment that is appropriate to the task and the level of risk.

General Medical fitness is prerequisite for the persons performing all the job at height.

In additions the following must be done:

- Hypertension / High blood pressure
- Epilepsy fits (Past History of illness)
- Vertigo/Height Phobia

Employees (including Contractors) who have not undergone additional medical check-up, required for 'Work at Height' **must not be deployed** on height work.

Medical tests performed for Work at Height is valid for in six months.

Refer to the GFL "Work at Height" Standard for details of mandatory requirements and safeguards of working at Height.

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4.2 Hot Work:

Few Examples of 'Hot Work' - the use or operation of any of the following:

- Welding or flame cutting equipment
- Grinder, electric drill or other non-flameproof electrical equipment
- Hammering, chipping, drilling (particularly in the presence of flammable/ combustible/ explosive atmospheres)
- Spark ignition engines or vehicle movement in plant area or non-approved compression ignition engines.
- Match, naked flame, cigarette lighter, flame or spark
- Hand tools that may create a spark
- Non-flameproof electric cables, cords, switches, lights, connectors or fittings
- Any device, which is likely to produce a source of ignition or be an ignition source, e.g. cell phones, pagers, radios, etc.
- Hot Work shall not be carried out in or adjacent to any hazardous area unless a Gas Test has been carried out using an approved explosimeter/toxic gas meter.
- The area or site shall be certified free of flammable vapors and gases and a "Hot Work Permit" issued by an authorized GFL signatory.
- A gas/vapor barrier fence may be required in some circumstances to exclude vapors from the work area.
- A fully charged fire extinguisher (pressure needle healthy sign) must always be stationed within easy reach in the work area where hot work is involved.
- Whenever any hot work is carried out in a highly flammable area then the presence of trained fire watch man/stand bay person is must, with necessary accessories near the spot where hot job to be carried out.
- P-JSA is required to be carried out for all work conducted as Hot work.

Refer to the GFL "Hot Work" Standard for details of mandatory requirements

4.3 CONFINED SPACE ENTRY:

A Confined Space:

- Is not intended as a regular workplace;
- Has restricted means for entry and exit;
- May have inadequate ventilation and/or an atmosphere which is either contaminated or oxygen- deficient; and is at atmospheric pressure during occupancy.

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- Confined Spaces include, but are not limited to: - tanks, vessels, vaults, certain pipelines, certain pits, elevated water towers, boilers, columns, reactors, spheres, silos, sumps, sewers, underground service and column skirts (those skirts where operator goes for valve operations is excluded from the definition, provided there is cross ventilation and there are two opening for ingress and egress.)
 - Note that these spaces could reasonably have atmospheres that are oxygen deficient or contain other hazards, typical examples of confined space hazards.
 - These spaces are subjected to potential hazards, viz., accumulation of toxic or flammable gas, vapors, inert gas, oxygen deficient (<19.5%v/v), flooding of liquid, steam or any other hazardous substance or development of hazardous conditions.
 - A Work permit must be issued before a person enters a Confined Space, and before the issue of the Work permit, a formal documented **P-JSA** must take place.
 - The Work permit must specify details of the location and the type of work to be done, with the P-JSA attached or cross referenced, and with the necessary protective measures detailed or attached.
 - Use of Destress meter is also mandatory when person entering inside the confined space.
- 4.3.1 Entry into a Confined Space must be in accordance with GFL Standard and a requirement demanded by legislation. Thus:**

No person shall enter a Confined Space unless:

- Formal P-JSA has been completed.
- All product and service lines entering the space are disconnected, blanked and checked. Drawing for the same is to be attached with the permit.
- All mechanical equipment such as stirrers and heaters are positively disconnected and electrically locked off.
- *All appropriate operating and maintenance personnel are advised of the activity and the period for which it is approved.*
- *All electric connection shall be provided with ELCB.*
- *For jobs in confined spaces only 24 V Flame proof hand lamp to be used. If lighting is required for any job involving release of flammable vapors or work is carried out in a hazardous area, it is mandatory to use flameproof electrical equipment. Only flameproof 24 V electric hand lamp shall be used. Use of 230 V hand lamps inside confined space/vessel is prohibited.*
- *When a person is entering confined space, the receiver must keep minimum one trained attendants at the manhole or entry point*
- *Wherever possible, lifelines with safety harness should be used for entry into confined space*

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- *Job involving hot work or entry into confined space, positive isolation by insertion of slip blinds or physical disconnection and blinding live ends are necessary.*
- *Vessel entry, where no hot work is to be carried out, shall be permitted if combustible gases are up to 5 percent of lower explosive limit [LEL]. Entry with an air-supplied mask shall be permitted with LEL of up to 20 percent. The Oxygen level shall be at least 19.5 to 22.5 percent volume, and the concentration of toxic gases below the threshold limits.*
- *No cylinders of compressed gas, other than those used for self-contained breathing apparatus, shall be taken into a confined space.*

4.3.2 Emergency response plans/Rescue plan are in place and appropriate training is completed

- Air-Supplied breathing equipment is being worn.
- All personnel involved have been instructed, practiced and proven competent in its use.
- Attendance of all personnel engaged in vessel entry, their entry and exit times is to be recorded. No entry during lunch /dinner breaks and shift changeover is to be ensured.
- An observer attendant/Standby person carrying walkie talkie / in centric safe mobile (IS Mobile) is stationed immediately outside the entry to the confined space all the time a person is inside.
- In case of Emergency attendant/Standby person should not go inside confine space. Only ERT members/ Control room team should be communicated for rescue.
- A minimum of a single self-contained BA set is on hand for emergency use, unless a P-JSA indicates additional units are required for higher risk activities, e.g. entry into underground tanks where there is only a single access manhole.
- *Entrant must have a medical test performed with in last six months. Required parameters checkup are included in routine medical checkup.*

4.3.3 Rescue Plan:

Job Risk Assessment to be carried out for confined space jobs. If Initial Risk is very high, (scoring 8 or more), A rescue plan is to be made.

Since Dahej Site has well trained Fire Team, a well-equipped Rescue Tender and Fire Tenders, following steps are to followed for rescue plan.

- PTW/Permit issuer has to inform the HoD.
- Call Fire Team and explain the job and associated hazards.

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- Fire team member has to visit the site and discuss a rescue plan with plant team, considering different scenarios.
- Document the rescue plan.
- Fire team has to keep his team informed and prepare the rescue equipment ready (rescue plan is to be made, keeping the equipment available in the rescue tender in mind).

NOTE: Rescue team and Rescue Tender has to mobilized in case of actual requirement

Refer to the GFL “Confined Space Entry” Standard for details of mandatory requirements and safeguards of working safely in a confined space.

4.4 ELECTRIC WORK:

4.4.1 Electrical Work permit requirement.

This electrical work Permit is required to be taken for any maintenance, installation & repair related work to be carried out on electrical equipment and power operated equipment in the electrical safety district/sub district/panel room at GFL, Dahej Site.

- Electrical work Permit is taken by all persons working on live or electrical equipment at the following locations inside the GFL Dahej Complex:
- Electrical Switch gears, Electrical switch yards, Electrical substation.
- Cable cellars, cable racks & cable trenches inside Substation or switchyard area.
- Transformers and Circuit breakers, UPS System, Battery /Battery charger systems inside Substation or switchyard area.
- Motors/ Motor Terminal Boxes, Power operated equipment inside Substation or switchyard area.
- Actuators of motorized valves, Junction Boxes, MCC Panels, Control Stations, HVAC systems, Cathodic protection systems, Electrical Heat Tracing, Electrical Earthing inside Substation or switchyard area.

4.4.2 Work permit requirement for Electrical activities.

The following examples are considered Electrical Work that required permit (for the purpose of this Standard):

- The task of adding a de-energized insulated jumper to an energized terminal (120VAC or less) while wearing proper PPE.
- Similar task of removing an insulated jumpers from an energized terminal (120VAC or less), if the removed jumper is de-energized as it is unplugged.
- *Use of approved fuse pullers and test equipment such as voltmeters, clip-on ammeters, hot sticks, and non-contacting voltage sensing devices for higher voltages will be considered as Hazardous work.*
- Normal operation of electrical welding machines.

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- *Testing of energized electrical circuits with approved test equipment when it can only be done without interrupting the circuit.*
- *Replacing electrical fittings (lamps etc.) when circuits are energized.*
- *Disconnecting motors without switching off motor starter.*
- *Fuse pulling and installing on energized circuits rated 415 volts and above (using approved tools)*
- Working on energized "draw out" type circuit breakers on 415 volts or higher.
- Installing and removal of insulating barrier when energized circuits are nearby and beyond the distances prescribed for Electrical Proximity Work.
- Installation of safety grounds when energized circuits are nearby and beyond the distances prescribed for Electrical Proximity Work.
- Work on high energy "battery system" of greater than 100 Vdc.
- All non-routine activities like Preventive Maintenance on electrical equipment.

The following are examples of Electrical Work on energized equipment performed by GFL personnel that may not require a work permit.

However, similar work performed by contractors require a work permit

- Changing out fluorescent lamp and bulbs with lighting fixture de-energized.
- Racking out / Rack-in motor starter and circuit breaker following other approved work procedures.
- Polarity testing of power point in non-hazardous areas.
- Working on electrical equipment where the maximum operating voltage is 50 Vac-gnd. or 100 Vdc.
- Testing of energized circuits with approved test equipment when it can only be done without interrupting the circuit in non-hazardous areas.
- *Racking out breakers for maintenance work on non-electrical portion of equipment less than 415 Vac*
- Troubleshooting of energized control circuits / relays by keeping power circuit de-energized in Electrical district ownership areas using approved test equipment and applying proper PPE.
- Troubleshooting energized control circuit in MCC/Electrical panel Room using approved test equipment, applying proper PPE and applying a work plan.
- Measurement of Current in LT feeders using calibrated measuring instruments by using defined PPEs as per approved procedures.
- Daily Temperature monitoring of LT critical panels using calibrated measuring instruments by wearing defined PPEs as per approved procedures.

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Refer to the GFL "Electrical Safety Management" Standard for details of mandatory Requirements.

4.5. LINE BREAKING:

Line breaking refers to the controlled opening of piping, ductwork, or containment systems that hold hazardous materials. This activity is considered high-risk due to the potential release of chemicals, gases, or steam, which can pose serious hazards to personnel and the environment.

This procedure applies to lines containing high pressure, high vacuum, corrosive, flammable, toxic, or hazardous fluids. Water supply, Process water, steam below 3 Kg/cm² and airline systems are excluded from line-breaking permits.

First breaking/opening of the line must be done in presence of filed operator/process engineer.

Examples of Line Breaking Activities

- *Removing bolts(even single bolt) from flanges*
- *Opening valves to release material*
- *Removing valve bonnets and check valves*
- *Turning spectacle plates (blanks/blinds)*
- *Disconnecting tubing or hoses*
- *Penetrating a line by mechanical means*
- *Opening inspection ports or sight glasses*
- *Replacing gland packing on valves*
- *Making the first cut on isolated pipelines*

Approver I & Approver II requirements:

- *PTW line breaking activities involving the following hazardous materials (but not limited to) require approval I & Approval II signature on PTW.*

1. *Sulfuric Acid*
2. *Chlorine*
3. *Methanol*
4. *Natural Gas*
5. *Caustic Soda*
6. *Hydrochloric Acid*
7. *Anhydrous Hydrofluoric Acid*
8. *Diethyl Ether (DHF)*
9. *Chloroform*
10. *Chlorodifluoromethane (R-22)*

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11. Tetrafluoroethylene (TFE)
12. Ammonia
13. Nitric Acid
14. Methyl Chloride
15. Methylene Dichloride (MDC)
16. Carbon Tetrachloride (CTC)
17. Florine gas

For all other line breaking activities, only approver 1 signature is required and PTW will be issue as per below Table. The permit issuer to decide the job execution team.

Ownership of line passing through multiple district:

Condition	Line Layout Description	Line Owner
1	Single supplier to single receiver.	Supplier Plant/Utility plant
2	Single supplier to multiple receivers.	Supplier plant/Utility plant
3	Multiple suppliers to single receiver.	Receiver/Utility plant

Note: In reference of above additional approval sign must be taken from district owner.

Responsibilities of Line Owner

- Lockout or Isolation: Ensure isolation from supplier and receiver sides, tagging and locking.
- Hazard Identification: Assess toxicity, flammability, pressure, temperature, and secondary energy sources.
- Draining and Decontamination: Drain low points, vent high points, flush and purge lines.
- Tagging and Isolation: Tag valves and blinds; keep valves open to avoid trapping material.
- Permit Coordination: Issue permit from safety district owner.
- Job Turnover: Review job scope before and after work.

System Preparation (Clearing & Verification)

- Drain low points and vent high points
- Flush or purge lines until hazards are reduced
- Verify cleanliness and absence of trapped pressure
- Stop work if hazards persist

Job Planning Details

- Hazard list and isolation points
- Flushing and clearing steps

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- PPE requirements
- Standby person duties
- Contingency plans
- Control of entry

Personal Protective Equipment (PPE)

Minimum : safety shoes, helmet, splash goggles, face shield.

Chemical/Corrosive/Toxic : chemical-resistant suit, gloves, boots, SCBA if required.

Flammable/Explosive : flame-resistant suit, nitrile gloves, SCBA as needed.

Steam/Nitrogen/Gas : face shield, leather gloves, SCBA for oxygen-deficient areas.

Job Execution

- Barricade work area
- Verify safety shower operability
- Perform initial break away from line of fire
- Loosen bottom bolts slowly
- Use secondary containment
- Stop and consult operations if plugs or pressure are found

Box-up and Return to Service

- Reassemble with correct bolts and gaskets
- Inspect integrity and cleanliness
- Remove isolation as per PTW closure
- Conduct turnover review and document sign-offs

Refer to the GFL “Line Breaking” Standard & Site procedure for details.

4.6 EXCAVATION ACTIVITIES:

- For any excavation activity 30 cm or deeper, obtain signatures from Electrical, Instrumentation, Utility, and HSE departments to confirm the absence of underground utilities in the marked area.
- A one-time approval is required for the same excavation area, and the approved permit must be attached to the next day's permit.
- If any underground utility is present in marked area, then concern dept. must written in permit & confirm manual / Mechanical Excavation on permit.
- Excavation area should be free from fumes/gases heavier than air because it may cause suffocation or poisoning.
- All excavation of depth more than 1.2 m or more should be treated as confined space and permit for entry into confined space should be obtained.

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- Excavation to a depth of 1 m or more shall be cut stepped back to give a suitable slope or securely held by bracing so as to avoid the danger of side to collapse.
- Barricading of 1 m height (with red and white band/ self-glowing caution boards) should be carried out for excavation beyond 1.5 m depth.
- Excavated earth shall not be dumped within 1 m of the edge of the excavation or depth the trench whichever is more.
- No vehicle should be allowed to go too close to the edge of the excavation because it may cause side collapse. Maintain at least 2 m distances from the edge of the excavation. Adequate and well-anchored stop blocks should be provided on the surface to vehicle being driven reversed into the excavation.
- No mechanical excavation is to be carried out up to a depth of 1.5 m from the ground level in operating areas. In such cases manual excavation is required to be carried out up to a depth of 1.5 m from the ground level.

Note -Test pits of 01 M deep are required to be excavated before actual excavation is started and all checking to be verified from concern authority prior to permit approval.

4.7. RADIOGRAPHY JOB REQUIREMENTS AND PERMITS:

- The radiography should preferably be carried out at a time, when occupancy is least.
- The supervisor must be available at site during the radiography work.
- Although radiography jobs do not qualify as hot jobs by definition, they still require a hot work permit due to the involvement of radioactive sources.

Before issuing permitting to start Radiography, ensure:

- TLD badge should be available with Radiographer
- Availability of Dosimeter and Radiation level Survey meter and its type
- Signage display
- Area barricading according to safe distance required
- Warning signal
- Information to surrounding plant and no other activities allowed in surrounding area.

Work Involving Radioactive Sources:

- The isolation or restoration of a radioactive source, including the use of mobile X-ray/radiography equipment.
- For radiography jobs on pipelines or equipment, the isolation and restoration of the radioactive source are essential.
- A separate radiation permit must be issued for works related to radiography.
- **Type of Radiation:**

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- X-Ray & Gamma Ray: Ir-192, Co-60
- Number of Sources: _____ Radiation Activity (Intensity): _____ Specify intensity in kV/Ci.

Measurement and Control of Radiation Exposure:

1. Collimator
2. Radiation survey meter
3. Film badge
4. Concrete wall barrier
5. Pocket dosimeter
6. Alarm/Lamp

Specify Barricade Off Distance: _____

4.8 RIGGING JOBS:

- **Pre-Job Meeting:** Conduct a meeting with all personnel involved in the rigging job to discuss the plan, potential hazards, and emergency procedures
- **Equipment Inspection:** Inspect all rigging equipment, including cranes, slings, and hooks, to ensure they are in good working condition and meet safety standards
- **Execution of Work:** Perform the rigging job as per the approved plan, ensuring all safety measures are followed. Continuous monitoring is essential to address any issues that arise
- **Post-Job Review:** After completing the job, conduct a review to identify any incidents or near-misses and document lessons learned for future reference.
- **Rigging Plan must be approved** as per checklist HSE/SAF/F/27
- i) For load ≤ 15 MT rigging plan should be approval by Maintenance Engineer, Maintenance Manager and Maintenance HOD.
- ii) For load > 15 MT rigging plan should be approval by Maintenance Engineer, Maintenance Manager, Maintenance HOD, and CES / Project head.
- iii) For load > 30 MT rigging plan should be approval by Maintenance Engineer, Maintenance Manager, Maintenance HOD, CES / Project head and Engineering Head.

4.9 HYDRAULIC AND PNEUMATIC SYSTEM WORK:

- Any machine & equipment where hydraulic energy like hydraulic jaw, hydraulic doors, chambers, cutters, hammers, pusher and lifting hydraulic tools where pneumatic energy is stored & used for operating, functioning of the machine.

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- Many hydraulic & pneumatic operated equipment working require a full hazard analysis during maintenance & repair as store hydraulic/pneumatic energy of the equipment having high potential hazard for accident if release suddenly, so release the hydraulic pressure or pneumatic pressure of related in an operating system prior to removing or adjusting components.
- The hot fluid in the system is under extreme pressure. If a worker is exposed to hydraulic fluid while removing or adjusting components, he or she may sustain burns, bruises, cuts, abrasions, or injection of fluid into the skin.
 - Do not rely solely on the hydraulic lift if you must work on hydraulic components with the system raised. Set the working unit on blocks as a precautionary measure.

5. ROAD CLOSER:

- The Respective plant HOD has to plan the road closure and inform to the HSEF, Security, SSM and Logistics teams via email at least 24 hours in advance.
- The Respective plant HOD & engineer Incharge decide on the road/street to be blocked and alternate route to be use.
- The Respective plant HOD has to marked (Road Blocked & alternate route) on the plant layout and share to the HSEF, Security, SSM and Logistics teams via email at least 24 hours in advance.
- The Respective plant HOD must obtain road block approval via email from the HSEF and Security teams for the period between 09:00 AM and 05:30 PM. For the period between 05:30 PM and 09:00 AM, approval must be obtained from the Deputy Unit Head or Unit Head.
- The Security and Logistics teams have to manage traffic and transportation according to the road block and alternate route decided.
- SSM must mention the road closure in the daily report for management's information every day until the road is clear.

6. FOR FIRE WATER NETWORK ISOLATION

Approval from the HSE department is required, with the knowledge of the Unit Head and Deputy Unit Head.

7. Changes in Conditions & Emergency Action:

- In the event of any of the following circumstances occurring during the course of an activity under permit, the Permit Holder must refer back to the Permit Issuer (through Permit Acceptor) on the appropriate course of action:
- The specified equipment or tools to be used not being available.

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- The type & nature of work changes.
- The process conditions change (e.g., unexpected release of gas from a nearby tank)
- An emergency in the area.
- Weather conditions change.

8. ROLES & RESPONSIBILITIES:

Table 1: Summary of appropriate people to fill each role

	Can Be a contractor	Issue Permits to others	Self-Permit?	Appoint Permit Issuers	Stop Work on site if necessary	Audit System
Unit Head	No	No	No	Yes	Yes	Yes
Permit Approver	No	No	No	Yes	Yes	Yes
Permit Issuer	No	Yes	No	No	Yes	No
Permit Acceptor	No	No	No	No	Yes	No
Permit Holder	Yes	No	No	No	Yes	No
Area Operator	Yes	No	No	No	Yes	No

Table 2: Roles and responsibilities in PTW

Sr No.	Name / Designation	PTW System Manager	Permit Approver	Permit Issuer	Permit Acceptor	Permit Holder	Area Operator
1	Unit Head/Dy Unit Head	Yes	Yes				
2	Plant Manager (HOD Operations)/Functional Head		Yes				
3	Maintenance Manager (HOD)		Yes	Yes			
4	Head, Project	Yes	Yes				
5	Project Managers		Yes				
6	Construction Manager/Engineer			Yes			
7	Shift In-Charge		Yes	Yes			
8	Shift Engineer (Operation/Maintenance)				Yes	Yes	Yes
9	Shift Operator						Yes
10	Maintenance Supervisor				Yes	Yes	
11	Plant Technician				Yes	Yes	
12	Maintenance Contractor				Yes*	Yes	

* In Project Areas/Activities only

Set out below are the principal roles and associated responsibilities required in operating a Permit to Work system.

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8.1 Unit Head/Project Manager: Refer PTW Corporator Standard.

8.2 Permit Issuer:

- The Permit Issuer (PI) is responsible for writing the permit for his/her safety district/Sub district/operating equipment/owner of equipment and for identifying (in conjunction with the person who is to undertake work/contractor if practical) hazards, risks and controls required to manage the risks.
- He must confirm that the hazards have been correctly identified and P-JSA and, that the proposed controls are adequate. The permit issuer has the requisite training, experience and competence in the PTW system.
- A permit Issuer can only be authorized to issue Permits in areas where they are trained to do so, e.g. Toxic Gas/oxygen Testing, Confined Space Entry, Hot Work, Working at Height etc.

The Permit Issuer is responsible for the following:

- Confirming that all hazards associated with the work have been identified, assessed, and managed and all relevant parties have provided their input and agree with the precautions.
- Ensuring that the permit accurately details the scope of works and that the documented controls combined are adequate to ensure the work can be completed in a safe manner.
- Ensuring that all necessary supporting permits/certificates, Checklist, P-JSA and drawings are attached or referred to in the permit.
- Ensuring that the Permit Holder knows the exact location of the work, all the hazards that may be present and all the precautions and controls that must be taken during the work
- Ensuring that the Permit Holder understands all the hazards and controls, will explain them to the work group, and will ensure they are maintained throughout the work.
- If discussion at the work site and checking the controls are delegated to another member of the operating team, the Permit Issuer still takes full responsibility for authorizing the work to go ahead.
- Confirming that all operational preparations, including process clearance and isolations, have been completed.
- Confirming that the work site is safe for the work to proceed.
- Specifying on the permit any additional precautions that need to be taken during the work.
- Identifying any conflicts between the proposed work and other activities in the area, and if necessary, cross-referencing the permits.

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- Informing all people who may be affected by the work.
- Knowing the status of the work at all times,
- Signing off and canceling the permit when the work is complete or suspended.
- The PI can only delegate their authority during a period of absence. Authority can only be delegated to another PI who satisfies the competencies required for a PI, as outlined in
- The PI must notify the custodian if their responsibilities have been delegated.

8.3 Permit Acceptor (Person who is responsible to execute work)

- The permit acceptor is normally Company staff but in certain instances (e.g., in a project site) may be the contractor.
- There are many activities that will be undertaken where the contractor is more of an expert than Company staff.
- It is not the purpose of the PTW system, or the Work Permit, to advise the contractor how to specifically address the expertise issues in their scope of work – the contractor will address this via completion of the P-JSA.
- The permit, when combined with a P-JSA, and contractor expertise, ensures that EHS risks involved in the work are addressed.
- Permit acceptor shall be responsible for executing the entire work in full compliance with the permit requirements.

His/her responsibilities include the following:

- Give tool box talk to permit holder and manpower involve in work. Mention their name in Permit and take signature.
- Understanding the potential hazards
- Understanding how the hazards are controlled.
- Knowing the emergency situations likely to occur in the work area and actions to be taken including evacuation procedure.
- Ensuring that the permit holder understands all the hazards and controls, will explain them to the work group, and will ensure they are maintained throughout the work.
- Informing all people who may be directly affected by the work.
- Signing and accepting the permit.
- Knowing the status of the work at all times,
- Transferring duties formally

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8.4 Permit Holder - Worker(s) who carryout Work

- The permit holder(s) shall be responsible for commencing and completing the authorized task while fully complying with all the requirements as listed in the permit. **His/Their responsibilities include the following:**
- Knowing and understanding the potential hazards.
- Knowing how to use PPE and complies.
- Knowing and understanding how the hazards are controlled.
- Knowing the emergency situations likely to occur in the work area and actions to be taken including evacuating the area.
- Knowing the terms and conditions of safe work

8.5 Permit Approver I/II/III

- The Permit Approver shall be responsible for ensuring that the hazards have been evaluated, adequate controls are provided and precautions are specified in the permit. He/she is responsible for the following:
- Knowing the hazards including potential hazards and confirming that all hazards associated with the work have been identified, assessed, and managed and all relevant agencies have provided their input and agree with the precautions.
- Knowing how to control and manage hazards.
- Verifying that the work area is safe to commence and carryout work, including checking that representative atmospheric sampling results are within the acceptable limits.
- Confirming that the permit accurately details the scope of works and that the documented controls are adequate to ensure the work can be undertaken and completed in a safe manner.
- Confirming that all necessary supporting permits, P-JSA are attached or referred to in the permit
- Confirming that the permit holder has been explained the exact location of the work, all the hazards & risks that may be present and all the precautions and controls that must be taken during the work.
- Confirming that the permit holder understands all the hazards and controls.
- Confirming that all operational preparations, including process clearance and isolations, have been completed.
- Signing the permit

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8.6 Area Operator

- Area Operator shall be responsible for ensuring that the work location has all the necessary safeguards available in line with the permit requirements.

His/her responsibilities include the following:

- Understanding the potential hazards in the work to be done and no new hazards are introduced during the execution of the assigned work. If the situation alters and new hazards occur or the safety requirements are not complied with, stop the work and inform the permit issuer.
- Understanding how the hazards are controlled.
- Knowing the emergency situations likely to occur in the work area and actions to be taken including evacuation procedure.
- Take appropriate actions when an emergency occurs.
- Ensuring that the permit holder complies with the controls and takes all specified precautions throughout the work.
- Ensure that the permit holder restores the site to its original condition after completion of work.
- Signing the permit.
- Knowing the status of the work at all times.
- Transferring duties formally.

8.7 Authorized Toxic Gas/Oxygen Level Tester

- The Authorized Toxic Gas/Oxygen Level Tester shall be suitably trained for the work he is undertaking and shall have access to and be competent to use approved and calibrated gas testing equipment.

The Authorized Toxic Gas/Oxygen Level Tester is responsible for:

- Carrying out gas tests (oxygen/ toxic gas and flammability) at the frequency specified on the permit and supporting certificates.
- Stopping the work and instructing personnel to withdraw from the area whenever a gas test or change in conditions indicates that it is unsafe for the work to continue, then reporting back to the Permit Issuer.

9. MONITOR, AUDIT & REVIEW of PTW SYSTEM

9.1 Records

- Permits to Work & supporting certificates shall be retained for a minimum period of 12 months for reference in case of an incident investigation or for auditing purposes. (Local legislation may specify a longer retention period).

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9.2 In-Depth Inspections

- In depth inspections should be carried out on a pre-determined frequency by persons independent of the PTW System (e.g. by line personnel from plants/areas other than where work is in progress (Plant HOD), SHE Personnel). The purpose of these inspections is to obtain an independent check on how the PTW process is operating.

9.3 Audits

- Each site shall audit compliance with this standard as part of its Safety audit program (this should include 1st and 2nd Party Audits).
- Independent, qualified technical auditors should carry out audits of the operation of the PTW System.
- These audits should be carried out as part of the EHS audit program at least annually for 1st Party Audits and 3 yearly for the 2nd Party Audits.

9.4 Deviation Process:

Ref CORPORATE HEALTH & SAFETY STANDARD -GFL/COR/EHSS 2002.00.

9.5 Loss of Work Permit Copy

- In case of lost permit, duplicate copy from permit issuer's copy may be used as original Permit for further processing.
- In such case Permit needs to be closed by concerned HoD.
- In case both copies are missing, permit shall be closed by giving a note, which shall be initiated by concerned HoD and must be approved by UH.
- **Training and Communication Requirements**
- Training is the responsibility of each plant HOD. In the event that interpretation or clarification is needed, questions shall be directed to the respective Unit Standards, Rules & Procedures Sub-Committee (SR&PSC), EHS Manager or the Corporate EHS.

9.6 Contact:

- The contact for this standard is the Standards, Rules & Procedure Sub-Committee (SR&PSC). Clarification/Interpretation regarding this standard shall be referred to the Chairman, Standards, Rules & Procedures Sub-Committee (SR&PSC).

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10. REFERENC:

1. Standard for Permit to work	GFL/COR/EHSS 2002.00 Rev 01
2. Job Safety Analysis (JSA)	GFL/COR/EHSS 2004.00 Rev 01
3. Work Permit format	Doc. No. HSE/SAF/02, 01/07/2021, (Rev.08)
4. Safety District Demarcation Layout	GFL-PP-DWG-SD-04-15-D0127 Rev. 02.08.2024
5. Standard Confine Space	GFL/COR/EHSS 2005.00
6. Standard Work at height and scaffolding	GFL/COR/EHSS 2001.00
7. Standard Hot Work	GFL/COR/EHSS 2011.00
8. SOP of Line breaking	GFL/COR/EHSS 3004.00
9. SOP of Rigging operation / crane safety	SAF/P/26 Rev. 00
10. Standard Excavation work	GFL/COR/EHSS 2007.00
11. SOP of Chemical Handling	SAF/P/29 Rev. 01
12. Electrical Safety Standard	GFL/COR/EHSS 2003.00
13. LOTO safety standard	GFL/COR/EHSS 2008.00
14. Indian Standards	IS 17893: 2023 JUN- BUREAU OF INDIAN STANDARDS

11 REVISION HISTORY:

Rev. No	Effective Date	Details of Changes	Reasons for Changes
0	01-11-2023	Procure updated as per PTW standard First Issue	New Standard GFL/COR/EHSS/2002.00
1	31.12.2024	Procure updated as per PTW standard with detail	Procedure written in detail
2	01.02.2025	Procure updated as per PTW standard and Site detail	Procedure update with clause and sub clauses
3	01.04.2026	1) Gas testing frequency change from 4 Hrs to 2 Hrs. 2) Minor changes done in Entry into a Confined Space, Electrical activities and Line Breaking 3) Electrical standard PTW included 4) 2 no's of format	1) New Corporate safety Standard electrical, Confined space & line breaking procedure updated. 2) Permit to Work corporate Safety revision Rev 01. 3) Corporate Electrical standard 4) Training records & Confined space entrance records added

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12 ANNEXURES:

List of Annexures as below

SAFETY PRECAUTIONS AND REQUIREMENTS FOR HOT WORK

The following safety precautions and requirements, though not necessarily exhaustive, shall be observed before and during the hot work:

1. The equipment to be worked on shall be positively isolated, both mechanically and electrically. Mechanical isolation shall be achieved either by disconnection, or by blanking following the isolation procedure.
2. Blanking is not required for stand-alone utilities supply piping. Utilities supply in this instance are: process water, potable water, LP/HP De-mineralized water, soft water, direct/indirect sea-water, ID cooling water, chilled water, fire water, LP/HP nitrogen, steam, plant air, process air, instrument air and steam condensate.
3. Prior to commencement of hot work on the utilities supply line mentioned above, it is of utmost important that the line to be worked on is verified to be cleared and depressurized. This verification can be done by: a) opening a bleed valve (drain valve); b) first break; or c) by doing a cold-cut. The drain valve should be left locked in open position throughout the duration of the hot work to prevent pressure build-up.
4. The equipment shall be freed of liquid and gas by depressurizing, draining, venting, purging and flushing.
5. In lined vessels or equipment containing pockets, recesses, double bulkheads, etc. precautions shall be taken to ensure that no material is trapped behind the lining. This may require cold drilling and steaming through behind the lining followed by gas testing. Where it is not possible to achieve gas free in the equipment or there is a doubt that all potentially hazardous materials have been removed, the space shall be inserted with nitrogen or water filled during hot work.
6. Any combustible material, including paper cartons, oily rugs and grass, located nearby shall be removed. Oil spills or deposits around the work site shall be cleaned up. Where it is not practical to remove combustible materials they shall be suitably covered or wetted with water.
7. Drain covers and surface manhole covers within 15 m from the work site shall be properly sealed throughout the work period to prevent emission of flammable vapors.
8. If flammable liquid is present in any open drain or ditch which runs within 15m of the work site, it shall be dammed at locations not less than 15m from the work site and pumped out
9. Any potential source of flammable gas or vapor emission, such as sampling point, vent or drain situated within 15m of the work site shall be rendered safe by sealing. If there is a

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potential release of flammable gas or vapor in the vicinity of hot work, it may be necessary to monitor the atmosphere using continuous gas monitors which will alarm on detection.

10. The equipment on which hot work is to be carried out must be certified as being gas free by the authorized gas tester. The equipment cannot be certified as gas free until all sludge, grease, scale, deposits, etc. have been removed, as subsequent work could disturb or heat the remaining materials and give rise to a hazardous condition.
11. Gas free condition must also be maintained within 15m from the work site throughout the hot work period.
12. If the hot work could result in the projection of material, the work site must be adequately contained. In the case of welding, cutting or grinding, where molten metal and sparks are produced, the work site must be adequately surrounded by fire resistant or flame retarding tarpaulin to contain such sources of ignition. Particular attention must be paid to elevated hot work where sparks and molten metal can fall and cause injury to passers-by or ignition to potential sources of flammable or combustible materials underneath. The immediate area around and underneath the hot work site must be properly barricaded or roped off with warning signs. Proper ventilation shall be used to ensure the hot work workers do not inhale the welding fumes.
13. All hot work sites must be equipped with sufficient portable firefighting appliances such as fire extinguishers of the appropriate type, fire hose lines charged with water from the nearby fire hydrants. The end sections of the 2½" hose line shall be 1½" hose connected to the gated "Y" piece with stream/fog shut-off nozzles for easy maneuverability.
14. The vicinity of the hot work area where there is a potential of sparks or molten metals falling off the ground shall be wetted with water where practicable.
15. The Performing and Approving Authorities must monitor the progress of hot work on a regular basis to ensure the work group abides by the permit conditions and requirements. In the case where hot work, especially heavy hot work, is carried out reasonably close to an operating plant of 'live' system containing hazardous materials, a 'fire watch/standby person' from the Performing Authority must be stationed at the work site to closely monitor and supervise the work where practicable.
16. The sitting of mobile equipment such as welding sets and generators must be outside the classified hazardous area and away from the pipe-bridge carrying flammable materials.
17. All electrical appliances, connectors and power sockets or extension cords to be used in the classified hazardous areas must be of industrial and flame-proof type.
18. Cigarette or tobacco smoking including the use of lighters or matches at the hot work site is STRICTLY PROHIBITED even if the work site is covered with a hot work permit.

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- **Prohibited Hot Work Areas**

1. PESO Tank farm area
2. Flammable chemical storage areas & process areas

SAFETY PRECAUTIONS FOR EXCAVATION WORK

The following safety precautions, though not necessarily exhaustive, shall be observed where applicable in the course of excavation work to prevent injury and property damage:

1. Before an attempt is made to excavate, the Performing Authority must know the exact locations of underground facilities.
2. No excavator or other digging machine shall be allowed to excavate close to underground facilities that must be left in place. A proximity limit for machine operations shall be established and the excavation completed by hand digging. When work has to be done over or around electrical cables, special precautions must be stated in the permit. If the underground cables, service tiles or any other piping are discovered in the course of excavation, the job must be suspended and the Approving Authority must be notified immediately. When mechanical excavator is used, a full-time contractor supervisor must be present at the site to guide the operator of the excavator and supervise the excavation. Close proximity digging with any machinery SHALL require a GFL person who is knowledgeable of the underground facilities.
3. Where a trench to be excavated exceeds 1.5m in depth, adequate piling, shoring and bracing shall be provided against the bank or side to prevent it from collapsing.

Piling, shoring and bracing used in a trench excavation to protect persons against falling or sliding material shall be of adequate strength. Planks used as sheet piling shall be at least 50mm thick. The maximum spacing between horizontal stringers or wales shall be such as to keep the planks within their safe bending stress. Shores and braces shall be of adequate dimensions for stiffness and shall be so placed as to be effective for their intended purposes. Each end of each stringer piece shall be separately braced.

Where trenching of more than 4 m in depth is done, such protection shall be constructed in accordance with the design and drawings of a professional engineer.

4. In every excavation exceeding 1.2m in depth, there shall be provided ladders, stairways or ramps to furnish safe access to and egress from such excavation.
5. Excavated material should be placed at least 0.60m back from the edge of the excavation, unless toe boards or other effective barricades have been installed to prevent fallback. The excavated material should be kept away from the drains and access ways.

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6. Open sides of an excavation where a person may fall in shall be guarded by adequate protective barricades and suitable warning signs shall be put up at conspicuous positions. If the trench is 3m or deeper, barricades made of rigid materials instead of ropes must be used. Covering of trench with canvas sheet or the like without such barricades is strictly prohibited.
7. Pick-and-shovel people working in excavations should be kept far enough apart to prevent injury to one another.
8. All road-crossing trenches must be covered by steel plates of sufficient strength for vehicles to pass through.
9. Vessel and Confined Space Entry Permit may be required before any person is allowed to enter a trench exceeding 1.5m in depth as the atmosphere in it may not be safe for entry. Special attention should be paid to those trenches being dug within process area or any place where there are possible sources of flammable or toxic gas emission. Such emission sources may be from the nearby equipment or the pipes underneath the trenches.

SAFETY PRECAUTIONS FOR COLD WORK

The following precautions (though not necessarily exhaustive) should be observed in relation to the performance of Cold Work:

1. For work involving the opening-up or de-energizing of equipment, the Lock, Tag and Try Procedure must be followed.
2. If equipment is to be opened up, it should first be depressurized, drained and purged of hazardous material under valve isolation, before positive mechanical isolation can be achieved.

The effectiveness of the valve isolation should be tested and if it is found that a valve is passing, appropriate measures will have to be adopted. These may include shutting additional valves or taking further equipment out of commission, the possibility of the presence of wax or ice under a valve should be considered, as subsequent melting would defeat the isolation. Where valve isolation is used, the valves should be locked off with chain and padlock or other equally effective device, to ensure that they are not opened inadvertently.

3. If toxic gases could be present, suitable breathing apparatus should be specified.

The possibility of the presence of pyrophoric material should be considered before admitting air. If necessary, the equipment should be water flushed/filled before opening up and wetted down afterwards.

4. Where hazardous materials such as hydrocarbons and chemicals are involved, mechanical isolation should consist of spading, blanking or disconnecting.

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An exception to this requirement would be some cases of minor work when locked off valve would suffice, providing the isolation is proved to be effective by opening drains on the equipment and proving those drains to be clear.

This exception is not made in the interests of expediency, but recognizes that swinging spades or making disconnection can be equally, or more hazardous, than some examples of minor work.

5. Work should not be attempted on any equipment where the possibility exists of hot material escaping, e.g., temperature exceeds its flash point, boiling point or auto-ignition temperature. The material should be allowed to cool before draining and extra care exercised when checking isolation integrity.

6. Particular care is required in achieving and proving isolation when equipment operates under a vacuum.

Wherever practicable, a formal maintenance preparation procedure should be written for any equipment where hazards such as hot material or vacuum operation are encountered.

7. Where driven machinery is to be worked on, the prime mover should be positively isolated and any switch gear locked off as per the Lock, Tag and Try Procedure.
8. Appropriate protective equipment must be specified.
9. The area around any work site must be appropriately identified and barricaded, if necessary, to prevent other personnel in or passing through the area from being exposed to hazards.

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Checklist for temporary Welding Machine/De watering pump & all other electrically operated portable tools

		TEMPORARY ELECTRICAL CONNECTION FORMAT	
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TITLE: Requisition Checklist for Temporary Electrical Connection of Welding Machine Sets / Dewatering Pump & all other Electrically operated portable tools			
Name of equipment Owner / Contractor's Supervisor / Contractor's Technician: -			
Plant :-	Name of Equipment :-	Date :-	
Area :-	Serial no. of Equip. :-	Checklist No :- 90098	
Location :-	Rated Current :-	Work Permit No :-	
To,	Required Phase :-		
Electrical Department,	Rating of ELCB/RCBO :-		
I request to connect / provide supply to (contractor) _____ in the premises of M/s. GFL, Dahej. I have personally checked the equipment and it is safe to my operating person and area. I owe the responsibility of maintaining good working conditions as well as Housekeeping in the working area and shall follow the safety norms decided by the GFL. Any of my worker, Labour or Supervisor found violating Safety rules of GFL, I shall be liable and responsible for the accident or damage of working persons, Machines, and plant.			
Requisitioner Name:		Sign:	Date & Time:
Points to be checked by Electrical Technician before connection of equipment.			
01. Size of Connecting Cable is proper and power cable is armoured /Shielded.	OK / Not OK		
02. Plug Top provided is properly rated & Plug top terminals are ok.	Yes / No		
03. Equipment is supplied through ELCB / RCCB /RCBO of 30mA.	Yes / No / NA		
04. The equipment is properly earthed to earthing point / strip and have no joints.	Yes / No		
05. Welding machine Insulation resistance value if validity expires (IR: _____).	OK / Not OK / NA		
06. Welding M/C Negative (return) lead shall be without joint & connected nearby job location.	Yes / No / NA		
07. Portable machines lead is properly rated and lead shall have healthy insulation.	Yes / No / NA		
08. Rotating portable hand tools shall have functional dead man switch.	Yes / No / NA		
09. Live parts at incoming of welding machine and Power tools are covered by shrouds.	Yes / No / NA		
10. Proper rated lugs and ferrules are used for connection of Portable machine.	Yes / No / NA		
11. The area where the job is to be done is well illuminated and spacious.	Yes / No / NA		
12. Portable machines are well protected against water and chemical ingress.	Yes / No / NA		
13. Portable machines / tools are Dry and Healthy (moisture free).	Yes / No / NA		
14. The motor being connected is properly checked. (IR value: _____)	Yes / No / NA		
15. The Starter is adequately rated and having the Relay range and setting as per guidelines.	Yes / No / NA		
16. The equipment being connected is properly earthed to earth point.	Yes / No / NA		
17. DC terminals of Welding machine are properly insulated and covered.	Yes / No / NA		
18. Extra holes and openings of equipment are sealed properly.	Yes / No / NA		
19. Extension board physical condition is ok.	Yes / No / NA		
20. Extension board ELCB/RCBO is working ok.	Yes / No / NA		
21. Cable rating, glanding & insulation condition is ok & it is armoured/shielded.	Yes / No / NA		
As all the above points have been checked by me, it is safe to operate the equipment. The owner / contractor of equipment is requested to maintain the same operating conditions throughout the job duration for safe usage of equipment.			
Sign of tech.		Sign of Engineer	
Name of Tech.		Name of Engineer	
Date & Time:		Date & Time:	
Extension: Extension of this Checklist shall be made if the job is continued after the initial validity of the work permit & job scope is not changed.			
Sign of tech.		Sign of Engineer	
Name of Tech.		Name of Engineer	
Date & Time:		Date & Time:	
- Note: This checklist is valid for 8 hrs unless extended. Maximum validity is up-to "C" shift end. - This checklist shall be considered as cancelled in case: flammable material leakage/emergency/ change of area / condition.			

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Checklist for Height Work Compliance:

 GFL GUJARAT FLUOROchemicals VALUE THROUGH GREEN CHEMISTRY	PTW: HEIGHT WORK COMPLIANCE		Doc. No. :	HSE/SAF/F/08
			Issue Date :	01.09.2024
			Rev. No :	00
PTW No. :		Job	:	
Date & Time :		Description	:	
SR. No.	CHECKPOINTS		STATUS (Yes / No / NA)	REMARKS
1	Is work being performed at heights above 1.8 meters using either a ladder or a scaffold?			
2	Are workers instructed on the proper method to wear and use the full body safety harness, and how to attach it to the lifeline or anchorage point?			
3	Are the full body safety harness, anchorage, lifeline, or fall arrest system inspected by a GFL authorized CES person before use by the workers?			
4	Is sufficient and secure anchorage or a lifeline provided?			
5	Are ISI marked, high-quality, and well-maintained full body safety harnesses with twin lanyards used while working at heights?			
6	Has a rescue plan been developed and implemented for work at height?			
7	When any work at height is to be carried out, is a safe means of access and egress provided?			
8	Is there adequate supervision to ensure that safe work practices for working at heights are in place			
IF LADDER IS USE FOR HEIGHT WORK				
8	Are the ladder and scaffold in good condition and marked with inspection tags?			
9	Is the ladder placed on a secure and level surface?			
10	Is the ladder placed at an angle greater than 75 degrees to the horizontal ground?			
11	When a ladder is used for access or as a working platform, are adequate handholds provided to a height of at least one meter above the landing point of the highest rung that a person's feet will reach?			
IF SCAFFOLDING IS USE FOR HEIGHT WORK				
12	Has the scaffolding been erected only on rigid/ firm / level surfaces?			
13	Has the scaffolding been designed to bear the intended load, and has a load calculation certificate been issued by CES?			

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14	Is a suitable platform with a toe guard, handrail, and mid-rail provided?		
15	Has the scaffold been inspected and certified by an authorized engineer before being put into use?		
16	Is provision made to arrange a duck ladder or crawling board for working on fragile roofs or asbestos?		
17	Has the scissor lift / man lifter used for work at height been inspected and certified by a competent person (CES person) before use?		
18	Are there any high-tension cables, honeybee combs, or other hazards directly above the work being performed at height?		

Authorization	Issuer	Acceptor	Area Operator
Name			
Signature			
Date & Time			

Note: (1) PTW-WAH to be done by issuer and acceptor jointly, if required include other cross functional members.

(2) Area Operator is responsible for all the control measures and the effective implementation by signing this "PTW-WAH" before start of the activity.

Checklist for Confined space Entry Compliance

 GFL GUJARAT FLUOROCHEMICALS VALUE THROUGH GREEN CHEMISTRY	PTW: CONFINED SPACE ENTRY COMPLIANCE				Doc. No.	:	HSE/SAF/F/07	
					Issue Date	:	01.09.2024	
					Rev. No	:	00	
PTW No.	:		Job	:				
Date & Time	:		Description	:				
SR. No.	CHECKPOINTS (Ref: CES Std GFL/COR/EHSS/2055.00)					STATUS (Yes / No / NA)		REMARKS
01	PROCEDURE FOR Working in CSE: Is the electrical isolation of rotating equipment done by removing the fuse, locking it, and tagging it?							
02	Are all connected pipelines, including those with CSE positively isolated?							
03	Are the ends of the detached lines fitted with blinds? The blinds should be indicated in the drawings/register should be maintained.							
04	Are isolation tags available on the valves to prevent accidental operation?							

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05	Is the draining, washing, cleaning, and purging done properly?	Done _____ by _____	
06	Is the CSE free from physical hazards like extreme cold or heat through proper cooling and air purging?		
07	Is there proper air ventilation inside the CSE, such as an exhaust fan or a movable blower provided with adequate capacity?		
08	Follow Electrical Safety: 8.1 Are only 24-volt bulbs with flameproof fittings used? 8.2 Are there any cable joints observed? 8.3 Is the power supply protected by an ELCB?		
09	Is the earthing resistance of the process vessel or tank less than 1 ohm? (Done with planning)		
10	Has the air quality been tested and recorded for the presence of toxic, poisonous, and flammable gases, as well as for oxygen percentage, at all locations of the vessel/confined space? and at different levels i.e. (top, middle and bottom) of vertical vessel, reactors		
11	Is there a provision for a properly sized ladder (Aluminium, M.S., or rope type) for ascending and descending inside the CSE, according to location requirements?		
12	Is a stretcher arranged at the nearest location of the CSE activity, if required?		
13	A distress meter (motion meter) is arranged, and anyone entering the CSE must wear it.		
14	Have the names and details of all personnel entering the confined space been recorded on the permit before entry?		
15	Arrangement of Fresh Air supplying facility (Airline respirator, Cascade system) available for respiratory protection against dangerous gases, fumes, vapours and oxygen deficient environment done? (Sulfuric Acid, Chlorine, Caustic, Natural Gas, Methanol, HCl, Anhydrous Hydrofluoric Acid, DHF, Chloroform, R22 (Chlorodifluoromethane), TFE (Tetra Fluoro ethylene), Ammonia, Nitric Acid, Methyl Chloride, MDC & CTC (carbon tetra chloride)).		
16	Have all jobs, inspections, and briefings on hazards and precautions for the confined space been completed for all personnel involved before starting the job?		
17	Does the person entering the CSE wear a full body harness with a double lifeline, anchored with a separate lifeline tied outside the CSE at a secure location?		
18	Standby Person Responsibility: Is there a person stationed outside the CSE near the		

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	manhole/opening as a standby, equipped with communication tools like a walkie-talkie for proper communication?		
19	Both the standby person and the person entering the CSE should be well-trained and possess sufficient knowledge about the process hazards and risks associated with that CSE		
20	Is a rescuer team member available during the CSE? Name: _____		
21	Is the rescuer team member trained in rescue operations?		
22	If the work continues for an extended period, does the person take a 15–20-minute rest intermittently. Is a gas test conducted after each rest and before re-entering the CSE & Recorded?		
23	Facilities Available at the location for Confined Space Entry Work: Multi-gas detectors with sensors for Oxygen, LEL, CO (carbon monoxide), H ₂ S (hydrogen sulfide), and portable Cl ₂ meters. 1. Online airline respirators located in different location of the plant. 2. SCBA sets provided in the plant control room and at key locations. 3. Trained first-aiders.		

Authorization	Issuer	Acceptor	Area Operator
Name			
Signature			
Date & Time			

Note: (1) PTW-CSE to be done by issuer and acceptor jointly, if required include other cross functional members.

(2) Area Operator is responsible for all the control measures and the effective implementation by signing this “PTW-CSE” before start of the activity.

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Checklist for Hot Work Procedure Compliance

 GFL GUJARAT FLUORO CHEMICALS VALUE THROUGH GREEN CHEMISTRY	PTW: HOT WORK PROCEDURE COMPLIANCE				Doc. No. :	HSE/SAF/F/06
					Issue Date :	01.09.2024
					Rev. No :	00
PTW No.	:		Job	:		
Date & Time	:		Description	:		
SR. No.	CHECKPOINTS				STATUS (Yes / No / NA)	REMARKS
A	PRE-WORK INSPECTION & PREPARATION:					
01	Is the area free of flammable or combustible materials?					
02	Is firefighting equipment, including fire extinguishers (ABC 9 Kg), readily available, sufficient, and operational near the work location.					
03	Are the involved equipment (e.g., welding machine, gas cutting set, tools) in good condition, with no damage or defects					
04	Is the LEL meter/multi-gas detector calibrated and functional? The normal LEL value of flammable vapor/ hazardous gases should be zero (Hazardous area) and never exceed 5%. LEL should be check within 2 hrs frequency. (In flameproof area)					
05	Has the gas cylinder checklist been approved by the acceptor, and are the cylinders kept in a trolley					
06	Are earthing connection tight at electrical extension box, machine, and welding objects?					
07	Is the mitigation plan for the task available for emergencies and communicated (ensure it is covered in the P-JSA)					
09	Is the welding machine on/off switch available near the work area to cut off power in case of an emergency?					
10	Is the welding machine frame grounded?					
11	Are welding or thermal cutting performing in confined spaces, gas cylinders and welding machines kept outside?					
B	TOOL BOX TALK (TBT):					
01	Gas lines purged individually before lighting the torch.					
02	Oxygen valve closed when lighting acetylene (DA Gas).					
03	The welder to avoid coiling or looping the welding electrode cable around parts of their body?					
C	AREA PREPARATION:					
01	Hot work area should be free from combustible material at least 25 ft radius distance.					
02	If the area cannot be free from combustibles, is it covered with fire-resistant materials (e.g., fire blanket)?					
03	Are workplace floors swept clean of combustible debris?					

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04	Are gaps or cracks properly sealed where sparks could enter and ignite hidden materials?		
D	DURING HOT WORK:		
01	Is a continuous fire watch assigned to monitor any signs of sparks or smouldering?		
02	Is clear access to exits and firefighting equipment maintained?		
03	Are torches lit by friction lighters or other approved devices only?		
E	POST-WORK PROTOCOL:		
01	Is the work area inspected and confirmed to be free of any signs of fire or potential ignition sources?		
02	Are the fire watch in place for at least 30 minutes after completing the hot work?		
03	Is hot work equipment and materials properly stored?		

Authorization	Issuer	Accepter	Area Operator
Name			
Signature			
Date & Time			

Note: (1) PTW-HW to be done by issuer and acceptor jointly, if required include other cross functional members.

(2) Area Operator is responsible for all the control measures and the effective implementation by signing this "PTW-HW" before start of the activity.

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Checklist for Line Break-Job Planning Sheet

	PTW: Line Break-Job Planning Sheet				Doc. No.	:	HSE/SAF/F/26
					Issue Date	:	01.09.2024
					Rev. No	:	00
PTW No.	:		Job Description	:			
Date & Time	:						
Job Planning sheet to be filled by shift in Charge before taking up of Line- break activity							
Identification of Hazards and Clearing method							
Hazards (Toxic/flammable/reactive/corrosive/high pressure / high temp etc.)		Clearing Method (Draining / flushing / purging / venting etc.)		Special precaution to be taken while clearing		PPE to be used (Also mention activities in bracket, during which it is not required)	
Duties of Standby Person:							
Contingency Plan / Instructions in case of single isolation device is used:							

Authorization	Shift In-Charge	Plant HOD
Name		
Signature		
Date & Time		

Note: (1) The P&ID / sketch of the system is attached showing points of isolation/blinding, proposed points for clearing and points of lockout.

(2) The above plan and information are discussed with all persons involved in the job.

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Checklist for Rigging Plan

	PTW: RIGGING PLAN			Doc. No.	:	HSE/SAF/F/27
				Issue Date	:	01.09.2024
				Rev. No	:	00
PTW No.	:		Job Description			
Date & Time	:					
Plant	:					
Location	:					
SR. No.	Details					
01	Equipment Detail (which is to be lifted)					
A.	Name					
B.	Dimension		a.	Weight:	MT	
			b.	Height:	Mtr.	
			c.	Length:	Mtr.	
02	lifting equipment detail (by which job will be lifted)					

Types	ID No. / Vehicle No	Capacity		last date of examination by competent person
		SWL (MT)	lifting length (meter)	

At this boom length and radius, as per load chart SWL= _____ MT

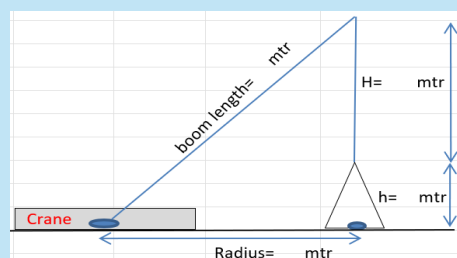
Total Weight (Job+ lifting hook+ slings) is to be lifted = _____ MT

Percentage of margin = Total Weight X 100 / SWL as per load chart = _____ %

It Should be:

75% of SWL of Crane Load Chart (When SLI and Wind velocity display not working)

85% of SWL of Crane Load Chart (When SLI and Wind velocity display is working)





GUJARAT FLUORO CHEMICALS LIMITED
Dahej Site



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Designation	Range of Load	Name	Approver Signature
Maintenance Engineer	All		
Maintenance Manager	All		
Maintenance HOD	All		

Note: Additional Approval Shall be taken in case of load more than 15 Ton.

Designation	Range of Load	Name	Approver Signature
CES / Project head	16 to 30 Ton		
Engineering Head	More than 30 Ton		

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PTW Format

Annexure-V

(Permit Format & Safety Precautions) HSE/SAF/02 , Rev-08

PERMIT VALIDITY: The validity of permit must be extended if work is allowed to be continued after it has expired. Work Permit No:

Section A: ☐ Cold Work ☐ Hot Work ☐ Confined Space ☐ Excavation ☐ Work at Height ☐ Line Breaking
☐ Rigging job ☐ Radiography ☐ Hydraulic/Pneumatic System ☐ Others _____

Section B: Job Details

Work Permit No.:

Plant/Unit: _____ Area/Location: _____ SAP Notification: _____

Job Description: _____ Equipment Name/ tag No: _____

Job Execution Agency: Own Employees / Contractor

Executive Department

Number of Persons at Work:

Section C: Nature of Work

☐ Work at Height ☐ Welding/Gas Cutting ☐ Hot Tapping ☐ Opening Line/Equipment ☐ Excavation ☐ Civil work
☐ Insulation ☐ Material Handling ☐ Painting ☐ Work on Electrical System ☐ Fire Network
☐ Mechanical Lockout ☐ Work on Fragile Roofs ☐ Closure of Roads ☐ Hydro-jetting
☐ Work on Inst. Systems ☐ Radiation Sources ☐ Grinding/Drilling/Cutting ☐ Online Sealing ☐ Others _____

Section D: Tools and Equipment To Be Used

☐ Welding Machine ☐ Gas Cylinders/ Gas Cutting Set ☐ Man Lift ☐ Non-Sparking Tools ☐ Mobile Crane/Winch
☐ Crane/ Farana/HEMM ☐ Lifting Tools & tackles ☐ Portable Electric Tools ☐ Scaffold ☐ Ladders
☐ Grinding/Cutting/Drilling Set ☐ Civil Hand Tools ☐ Hydraulic Tools ☐ Pneumatic Tools ☐ Lifeline
☐ Mechanical Handy Tools ☐ Electrical Insulated Tools ☐ Others _____

Section E: Hazard/Risk Considerations

☐ Fire / Flash ☐ Fall from Height ☐ Oxygen Deficiency ☐ Chemical/Toxic Gas Exposure
☐ Corrosive Mat. ☐ Hot Surface ☐ Stored Energy ☐ Crush/Cut Injury ☐ Pressurized Equip.
☐ Noise Exposure ☐ Dust Exposure ☐ Radiation Hazard ☐ Caught inside ☐ Electric Shock ☐ Slip/Trip
☐ Strike by ☐ Crush/cut Injury ☐ Spark ☐ Ergonomic ☐ Awkward Position ☐ Splash
☐ Splash ☐ Skin Burn ☐ Environment Exposure ☐ On/ Off Site Emergency ☐ Static Charge
☐ Exposure to rotating equipment ☐ Simultaneously operation ☐ Others _____

Section F: Job/Equipment Preparation

☐ Mechanical Isolation ☐ Equipment/vessel de-pressurized/drained
☐ Equipment/Pipe Blinded ☐ Use of 24 Volt Lamp (Flame-Proof)
☐ Isolation/Blinding Tag No. _____ ☐ Vehicle Entry Procedure Compliance

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☐ Grounding Electrical Equipment/Welding machine Compliance checklist	☐ Protecting the surrounding area from sparks/fire
☐ Rigging Plan	☐ Road Closure Required
☐ Crane/ Farana Certified	☐ Ladder (Type: _____)
☐ Height Work Compliance	☐ Hot Work Procedure Compliance
☐ Fire Blanket Booth	☐ Fire Fighting Team Required
☐ Confined Space Entry Procedure Compliance	☐ Drain/Sewer Cover
☐ Electrical Tool Checklist No. _____	☐ Area Barricading
☐ Scaffolding clearance By: _____ (Sign: _____)	☐ LOTO Compliance
☐ Excavation clearance By: Fire: _____ Electrical: _____ Instrument: _____ IT: _____ Utility: _____	☐ Lifting Tool/ Sling/ Hook Checked
☐ Stand-by Person required/ Fire watch: (Name: _____)	☐ Pre JSA Completed
☐ Others (Please specify): _____	☐ Rescue Plan
	☐ Cylinder Checklist No. _____
	☐ Atmosphere tested for oxygen, flammable and toxic gases
	☐ Isolation of Other Energy Source

Section G: PPE and Fire Protection

☐ Fire Extinguishers (Type: _____ No. _____)	☐ Fire water available/Hose ready	☐ Fire blanket	☐ Face Shield
☐ Safety Shoes	☐ PVC Gloves	☐ Cotton Gloves	☐ Gum Boot
☐ SCBA	☐ Dust mask	☐ Cartridge Gas Mask	☐ Eye Protection
☐ Safety Glasses	☐ Chemical Gloves	☐ Ear Plugs	☐ Ear Muffs
☐ Splash Goggles	☐ Rubber Gloves	☐ Protective suit	☐ Distress Alarm systems
☐ Turtle Suit for Hydro Jetting	☐ Others _____	☐ LEL Meter	☐ Leather Gloves

Section H: Isolation:

Electrical Energy

Other Energy Sources

I certify that the machine / equipment is isolated from source of electrical energy by Switching off / fuse removal / LOTO

Name _____ Signature _____

Date ____/____/____ At _____ Hrs

I certify that the machine / equipment is isolated from other source of energy

Name _____ Signature _____

Date ____/____/____ At _____ Hrs

Section I: Section I: Gas Test (for Hot Work and Confined Space Entry)

TWO-HOURLY GAS TEST, IF REQUIRED, SHALL BE CARRIED OUT BY THE APPROVED GAS TESTER AND RESULTS RECORDED ON REVERSE OF THIS FORM.

ATTACH A LIST OF ALL ENTRANTS INTO THE CONFINED SPACE.

Permit Valid From _____ Hrs To _____ Hrs Date _____ Unless extended as below

Permit Extended on Date _____ Till _____ Hrs Name _____ Signature _____

Section J: Permit Issuer - I certify that all the hazards/risks have been identified/assessed and controlled adequately (refer to JSA No. --)

Name _____ Date _____ At _____ Hrs

Signature _____

Section K: Permit Acceptor - I have read the precautions and safety measures mentioned herein and I agree to abide by the same. The safeguards have been implemented. Tool box talk given to persons working/impacted (refer the list of work executors).

Name _____ Date _____ At _____ Hrs

Signature _____

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Section L: Permit Approver – I have personally inspected and checked all requirements and approve the permit.

First Approver Name _____ Date _____ At _____ Hrs Signature _____

Second Approver Name _____ Date _____ At _____ Hrs Signature _____

Section M: Area Operator - I have been communicated and understood the work to be executed in the area, precautions and safety measures to be taken as mentioned herein.

Name _____ Date _____ At _____ Hrs
Signature _____

Section N: Permit Closure/Cancellation:

☐ Job Completed & Housekeeping restored

☐ Job to be Completed.

Please Issue a new Permit

☐ Permit is Cancelled and Closed

☐ Due to _____

Authority

Name

Signature

Date

Time

Power restored by

Acceptor

Area Operator

Issuer

Tick **all** appropriate boxes ✓

IMPORTANT: THE PERMIT IS AUTOMATICALLY SUSPENDED IN THE EVENT OF PLANT-WIDE EMERGENCY OR IF THERE ARE CHANGES IN WORK SITE CONDITIONS MAKING IT UNSAFE TO CONTINUE WORK.

Tool Box With WORKER LIST

NAME	AGENCY	Signature	NAME	AGENCY	Signature

PERMIT RENEWAL RECORD

	Date	Issuer	Acceptor	Area Operator	NDO/SSM (After 22:00 Hrs)	Approver
Name & Signature & Time						
Name & Signature & Time						
Name & Signature & Time						
Name & Signature & Time						

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TEMPORARY ELECTRICAL LOTO LIFTING FOR EQUIPMENT TRIAL

	Issuer Name, Sign & time	Accepter Name, Sign & time	Area Operator Name, Sign & time
Date:			
Date:			

GAS TEST RECORD

GAS METER TYPE: _____ DATE OF LAST CALIBRATION: _____

AUTHORIZED GAS TESTERS

NAME	FUNCTION	SIGNATURE

DATE	TIME (hrs)	CONCENTRATION OF FLAMMABLE GAS, %LEL				GAS TEST LOCATION	TESTED BY	SIGNATURE
DATE	TIME (hrs)	CONCENTRATION OF O ₂ and TOXIC GAS, PPM				GAS TEST LOCATION	TESTED BY	SIGNATURE
		O ₂	CO	H ₂ S	Other (e.g., Cl ₂ , H ₂ , CS ₂)			

CONFINED SPACE ENTRANCE RECORD

Equipment Tag No:

Name of Equipment

Sr no.	Name	Time in	Signature	Time Out	Signature
1					
2					
3					
4					
5					

Note: The permit issuer shall assess the type of confined space and decide the permitted stay time accordingly.

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Isolation tags

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PTW format for Electrical Work
Annexure
(Permit Format & Safety Precautions)

PERMIT VALIDITY: The validity of permit must be extended if work is allowed to be continued after it has expired.

Section A: TYPE OF PERMIT - ELECTRICAL

Section B: Job Details

Work Permit No:

Plant: _____ Area/Location: _____ SAP Notification: _____

Job Description: _____

Equipment Name/ tag No: _____ Job Execution Agency: _____

Executing Department: _____ Number of Persons at Work: _____

Section C: Nature of Work

- ☐ Work at Height ☐ Work near Live supply ☐ Hacksaw cutting ☐ Testing / Inspection
☐ Painting / letter writing ☐ Soldering ☐ Grinding / Drilling ☐ Welding ☐ Cable jointing
☐ Work on Energized System ☐ Work on De-energized system ☐ Cable laying ☐ Control Wiring
Work on LT Feeder ☐ WORK ON HT Feeder ☐ Motor Repair Work ☐ Other _____

Section D: Tools and Equipment To Be Used

- ☐ Welding Machine ☐ Man Lift ☐ Busbar Bending Tools ☐ Blower ☐ IR tester
☐ Crane/ Farana/HEMM ☐ Lifting Tools & tackles ☐ Scaffold ☐ Ladders
☐ Grinding/Drilling Set ☐ Hydraulic Tools ☐ Mechanical Hand Tools ☐ Electrical Insulated Tools ☐ Others _____

Section E: Hazard/Risk Considerations

- ☐ Fire / Flash ☐ Fall from Height ☐ Electrical Shock ☐ Hot Surface ☐ Stored Energy
☐ Crush/Cut Injury ☐ Noise Exposure ☐ Dust Exposure ☐ Arc flash ☐ Slip/Trip
☐ Strike by ☐ Awkward Position ☐ Static Charge ☐ Others _____

Section F: Job/Equipment Preparation

- ☐ Isolation / Switched OFF ☐ Approach Boundary Marked
☐ Absence of Voltage checked & terminals Discharged ☐ Escape route clear
☐ Power / Control Fuse removed ☐ Power Cable Disconnected
☐ Rigging Plan ☐ Adequate Ventilation provided
☐ Certified Ladder (Sr. No: _____) ☐ Area Barricaded / Caution Board

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- | | |
|---|--|
| ☐ Height Work Compliance | ☐ Lifting Tool/ Sling/ Hook Checked |
| ☐ Pre JSA done | ☐ Electrical Portable Tools certified |
| ☐ Insulation Mat / floor Coating Available | ☐ Breaker Racked out |
| ☐ Scaffolding Compliance | ☐ Hot Work Compliance |
| ☐ Stand-by Person Required: Name: _____ | ☐ Others (Please specify): _____ |

Section G: PPEs and Fire Protection

- ☐ Fire Extinguishers (Type: ____ No. ____)
- ☐ Electric shock proof Shoes ☐ Dust mask
- ☐ Electrical Insulated Gloves ____ KV ☐ Ear Plugs ☐ Ear Muffs ☐ safety hand gloves
- ☐ Protective FR suit ____ Cal/cm2 ☐ Safety Goggles ☐ Eye Protection / Arc flash Face shield
- ☐ Non-contact Voltage tester ☐ Multimeter ☐ Discharge Rod ☐ Rescue Rod ☐ Other _____

Section H: Isolation:

Electrical Energy

Other Energy Sources

I certify that the machine / equipment is isolated from source of electrical energy by Switching off / fuse removal / LOTO

(Tag no: _____)

Name _____

Signature _____

Date ____/____/____ At ____ Hrs

I certify that the machine / equipment is isolated from other source of energy (Tag no: _____)

Name _____

Signature _____

Date ____/____/____ At ____ Hrs

Permit Valid From ____ Hrs To ____ Hrs Date ____ Unless extended as below

Permit Extended on Date ____ Till ____ Hrs Name ____ Signature _____

Section I: Permit Issuer - I certify that all the hazards/risks have been identified/assessed and controlled adequately & Tool box talk given to persons working/impacted (refer the list of work executors). (refer to P-JSA No.)

Name ____ Date ____ At ____ Hrs Signature _____

IMPORTANT: THE PERMIT SHALL BE APPROVED BY APPROVER-I & APPROVER-II ONLY IN CASE OF WORK ON ELECTRICAL SYSTEM ABOVE 500V AC/DC INSIDE ELECTRICAL SAFETY DISTRICT UNLESS OTHERWISE SPECIFIED.

Section J: Permit Approver 1 – I have personally checked & verified all requirements.

Name ____ Date ____ At ____ Hrs Signature _____

Section K: Permit Approver 2 – I have personally checked & verified all requirements.

Name ____ Date ____ At ____ Hrs Signature _____

Section L: Plant Process Shift in-charge - I Agree to release the asset / Equipment for the work specified.

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Name _____ Date _____ At _____ Hrs Signature _____

Section M: Permit Acceptor (L5, L5s, L6) - I have been communicated and understood the work to be executed in the area, precautions and safety measures mentioned herein and I agree to abide by the same. The safeguards have been implemented.

Name _____ Date _____ At _____ Hrs Signature _____

Section N: Permit Closure/Cancellation:

☐ Job Completed & Housekeeping restored

☐ Job is incomplete.

Please Issue a new Permit

☐ Permit is Cancelled / rejected and Closed

Due to _____

Authority	Name	Signature	Date	Time
Acceptor				
Issuer				
Power restored by				

Tick **all** appropriate boxes ☐

IMPORTANT: THE PERMIT IS AUTOMATICALLY SUSPENDED IN THE EVENT OF PLANT-WIDE EMERGENCY OR IF THERE ARE CHANGES IN WORK SITE CONDITIONS MAKING IT UNSAFE TO CONTINUE WORK.

Tool Box with WORKER LIST

NAME	AGENCY	Signature	NAME	AGENCY	Signature

PERMIT RENEWAL RECORD

	Date	Issuer	Approver-1	Approver-2	Acceptor	Area Operator	Process Shift In-charge	Valid till
Name, Signature & Time								
Name, Signature & Time								

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GAS TEST RECORD

GAS METER TYPE: _____

DATE OF LAST CALIBRATION: _____

DATE	TIME (hrs)	%LEL	GAS TEST LOCATION	TESTED BY	SIGNATURE

Electrical PTW Proper PPE:

Document Number: GFL/COR/EHSS/2012.00																						
	Head, Face, Eyes, Ears					Respiratory			Body					Feet			Hands					
Location/ Activity	Safety Glasses	Welding or cutting Goggles	Chemical Splash Goggles	Face shield	Safety helmet	Hearing Protection	Half-face (mouth, nose)	Full-Face (air purifying via cartridge)	Dust Mask	Full Body protection (Jacket /apron (rubber, PVC))	Fire -Retardant Clothing 8cal/cm2 OR Arc Flash Suit with Hood (LV) 8cal/cm2	Arc Flash Suit with Hood (HV) 40cal/cm2	Fall protection – Safety harness	Gum Boots	Safety shoes	Di-electric rubber shoes	Cut resistance gloves	Chemical resistant gloves	High voltage gloves	Low voltage gloves(1.1KV)	Cotton Gloves (with or W/O PVC Dots)	Heat resistant gloves
LT LOTOTO-ACB	✓			✓	✓						✓					✓				✓		
LT LOTOTO-MCCB/SFU	✓			✓	✓						✓					✓				✓		
LT LOTOTO-DOL/SD Starter	✓			✓	✓						✓					✓				✓		
LT-Power Change Over	✓			✓	✓						✓					✓				✓		
HT LOTOTO	✓				✓							✓				✓			✓			
HT-Power Change Over	✓				✓							✓				✓			✓			
Feeder PM	✓			✓	✓											✓					✓	
Thermography(LV)	✓			✓	✓						✓					✓					✓	
Current / Voltage measurement (LV)	✓			✓	✓						✓					✓			✓			
Motor PM	✓				✓											✓					✓	
Transformer/Busduct PM	✓				✓								✓			✓					✓	
Earthing checking	✓				✓											✓					✓	
Transformer/Panel reading	✓				✓											✓					✓	
Light checking & repairing	✓				✓								✓			✓					✓	
Motor bearing replacement	✓				✓											✓					✓	
Relay reset	✓				✓											✓					✓	
SMF-Battery maintenance	✓				✓											✓					✓	
Ni-Cad Battery Maintenance	✓				✓											✓					✓	

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Motor LLF	✓				✓	✓			✓						✓					✓		
Control Circuit checking	✓				✓										✓					✓		
	Head, Face, Eyes, Ears					Respiratory			Body				Feet			Hands						
Location/ Activity	Safety Glasses	Welding or cutting Goggles	Chemical Splash Goggles	Face shield	Safety helmet	Hearing Protection	Half-face (mouth, nose)	Full-Face (air purifying via cartridge)	Dust Mask	Full Body protection (Jacket /apron (rubber, neoprene-suits)	Arc Flash Suit with Hood (LV)	Arc Flash Suit with Hood (HV)	Fall protection – Safety harness	Gum Boots	Safety shoes	Di-electric rubber shoes	Cut resistance gloves	Chemical resistant gloves	High voltage gloves	Low voltage gloves(1.1KV)	Cotton Gloves (with or W/O PVC Dots	Heat resistant gloves
UPS/DCDB/Battery Charger PM	✓				✓				✓							✓					✓	
ACB/MCCB Isolation	✓			✓	✓						✓					✓				✓		
SFU Maintenance (@ Live panel)	✓			✓	✓						✓					✓				✓		
Motor Greasing	✓				✓											✓					✓	
Panel Busbar removal/Installation	✓				✓											✓	✓					
Heater Maintenance	✓				✓				✓				✓			✓					✓	
AC Maintenance	✓				✓				✓				✓			✓					✓	
Panel Shifting & Erection	✓				✓											✓					✓	
Cable Tray Installation	✓				✓								✓			✓					✓	
Cable Laying/Dressing Work	✓				✓								✓			✓					✓	
Temporary Power Connection	✓				✓											✓					✓	
CA/Rectifier Current measurement	✓			✓	✓						✓					✓				✓		
CA/Harmonic filter bank maintenance	✓				✓								✓			✓			✓			
CA/Polarization Rectifier Maintenance	✓				✓						✓					✓				✓		
CA/ DC Isolator Maintenance	✓				✓											✓					✓	
R125 HF /Electrical Job	✓		✓		✓			✓		✓						✓		✓		✓		
CPP/66kV Line Isolation	✓				✓							✓				✓				✓		
CPP/Isolator or Earth switch operation	✓				✓							✓				✓				✓		
CPP/66Kv SF6 CB Operation	✓				✓							✓				✓				✓		
CPP/66Kv bay maintenance	✓				✓				✓							✓					✓	
CPP/Generator PM	✓				✓				✓							✓					✓	
CPP/ESP Field Discharge	✓				✓											✓			✓			

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PERMIT MATRIX FORMAT: Department/Plant: _____ Date: ____/____/____

Department	RESPONSIBILITY	COLD WORK PERMIT	HOT WORK PERMIT/ RADIOGRAPHY	CONFINED SPACE PERMIT/ EXCAVATION	WORK AT HIGHT PERMIT	LINE BREAKING
PRODUCTION	ISSUER					
MECHANICAL	ACCEPTOR					
ELECTRICAL	ACCEPTOR					
INSTRUMENT	ACCEPTOR					
CIVIL	ACCEPTOR					
SAFETY DISTRICT OWNER	Permit Approver I					
	Permit Approver II					
Gas testers						

HSEF Head Signature: _____ Name: _____ Date: ____/____/____

HOD Signature (For issuer, acceptor, gas tester): _____ Name: _____ Date: ____/____/____

Site Head (Only for Approver I & II Sign): _____ Date: ____/____/____

PLANT NAME _____ TRAINING RECORD

FUNCTION	NAME	TRAINING DATE	RETRAINING REQUIRED DATE
ISSUER			
ACCEPTOR			
Gas testers			

HOD Signature _____ Name: _____ Date: ____/____/____

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Emergency rescue plan for Confined space

EMERGENCY RESCUE PLAN	
DETAILS OF ACTIVITY	
RESCUE TASK	
DETAILS STEPS OF RESCUE	
PERSON RESPONSIBLE	
REQUIREMENT OF RESCUE EQUIPMENT	1. Lifeline rope. 2. Retractable fall arrester with Winch/Tripod/Double block pulley with rope. 3. Oxygen meter. 4. Intrinsically Safe Torch. 5. lock arrangement for manway cover as it is hinge type door. 6. Motion sensor (DSU) 7. Full body harness 8. Skid stretcher/Stretcher
EMERGENCY CONTACT DETAILS	
COMMUNICATION SYSTEM	PA System/I-Safe/Walkie Talkie Set
CHECKLIST OF RESCUE PLAN	Rescue pathway should be clear of any obstruction.
	All the rescue equipment has been checked & found ok
	All the rescue personnel are trained on rescue plan
	All the rescue equipment's are in place & accessible
	Mock drill/ demo of rescue plan has been carried out
	Area is restricted for unauthorized entry
	All concerned Depts/ sections / agencies are informed
	Back up arrangement is ready in case of power failure
	Communication equipment's are adequate & ok
	Rescue team members are medically fit
	Rescue team is dedicated for the job
	First aid procedure in place
	Gas testing equipment in place
	Breathing apparatus (MASK/SCBA/ SCBA TROLLEY) is ready
Rescue team members are aware about hazards	
Note:	
Rescue plan Prepared by: HSEF team	
Rescue plan to be reviewed by: HoD - Concerned Plant and HSEF Dept.	
Rescue plan to be Approved by: Plant Head (in his absence, the person officiating)	

Rescue plan Prepared By

Rescue Plan Reviewed by

Rescue Plan Approved By

Safety district map Dahej Site

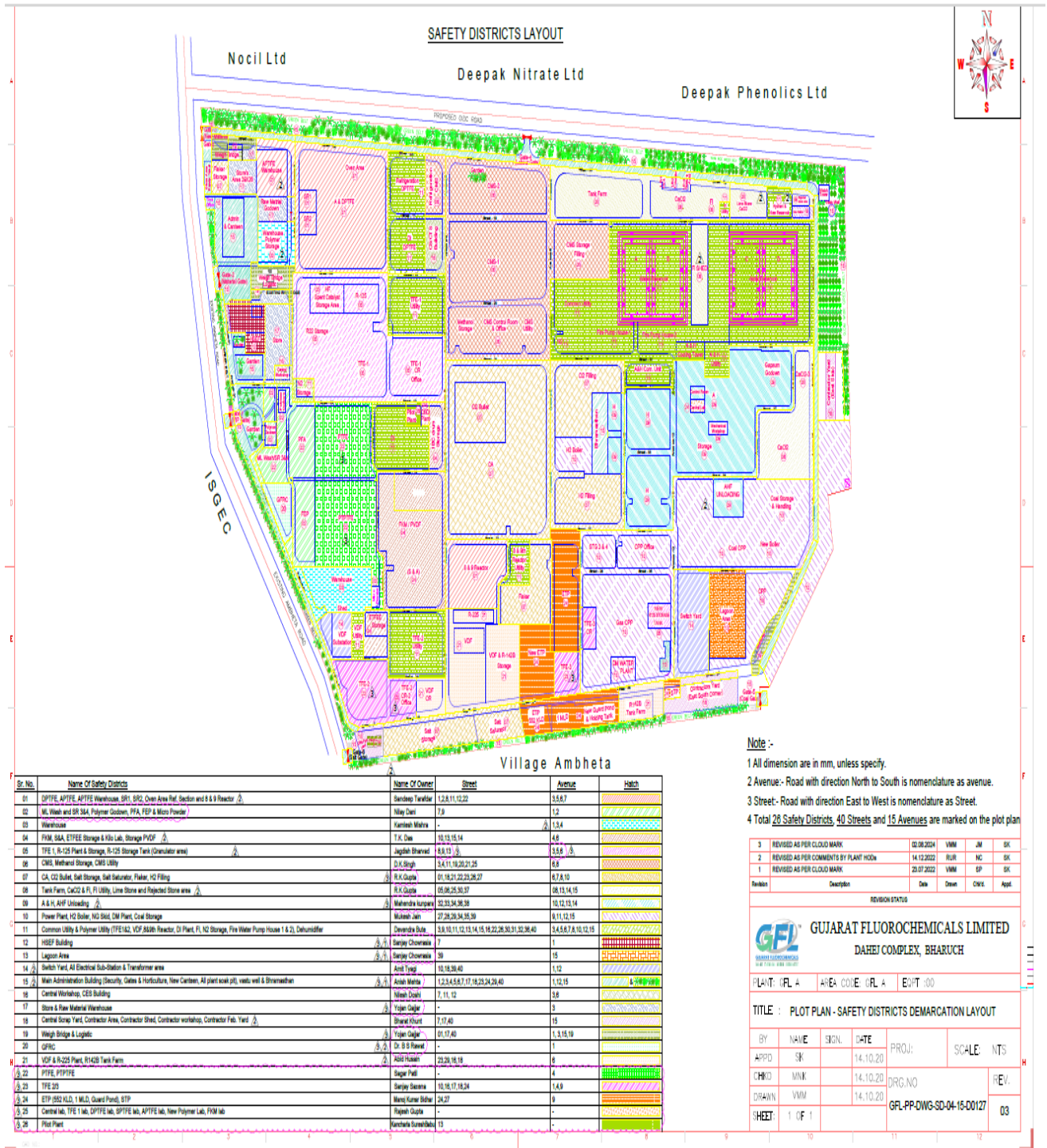
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Procedure for Safety District Owner

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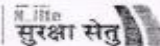
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STANDARD OPERATING PROCEDURE



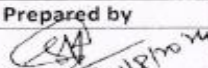
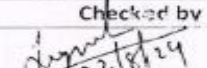
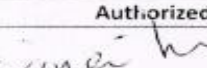
TITLE: PROCEDURE OF DISTRICT OWNER

Document Number: SAF/P/04

Issue Date: 14.08.2024

Rev No: 00

Page No: 01 to 03

	Prepared by	Checked by	Authorized by
Signature			
Name	Mr. Sanjay Chowrasia	Mr. Jignesh Parmar	Mr. Sanath Kumar
Designation	HSEF Head	Dy. Unit Head	EP (O)

TITLE : PROCEDURE OF AREA OWNER

01. PURPOSE:

The purpose of this guideline is to outline the Area Ownership concept. The objective of area ownership is to eliminate injuries through pro-active Safety dialogue with the people working at shop floor and to identify and eliminate unsafe conditions prevailing at shop floor, to uncover (and stop) unsafe acts and behaviors, reinforce safe behaviors, maintain standards, measure effects, increase Safety awareness, demonstrate leadership commitment to Safety, and improve Safety culture.

02. SCOPE:

This policy is applicable to all plants of GFL-DAHEJ.

03. Management Responsibilities:

Site Safety Apex committee will be responsible for implementing this guideline through line managers. EHS department is responsible for any revision in this procedure in consultation and approval of Unit Head / Dy Unit Head.

04. How does it work:

- ✓ The concept of Area Owner is as follows:
- ✓ Entire site (plant premises) is divided into a number of small areas known as "Safety District, Street, Avenue and Hatch"
- ✓ Names of the Safety Districts owner are given in Annexure-A. Please refer attached drawing for better clarity of jurisdiction.
- ✓ All the safety district owners shall constitute an area Owner who shall report to the district Owner.
- ✓ The overall accountability lies with the Area Owner.

05. Roles and Responsibilities of Area Owner:

- ✓ Shall be the owner of the facilities/areas.
- ✓ Shall have the responsibility of housekeeping, maintenance and upkeep of facilities.
- ✓ Shall have the overall control over the safety related activities in their districts.
- ✓ Shall ensure observance of the regulations and directions laid down in the company Health, Safety and Environmental Policy and procedures. He shall also ensure compliance to Statutory Regulations such as Factories Act and Rules, Petroleum/SMPV/Gas Cylinder Act and Rules, Environment Protection Act etc., though other department heads are responsible for the fulfillment of respective elements in their functional areas.
- ✓ Shall have the responsibility to sign the work permits by themselves or through their authorized personnel as per the procedure, safe work permit system. The Area Owner of HR, Administration will seek the advice of HSEF Head / Chief Manager - HSEF for issuing Hot work / Excavation work permit in their districts.
- ✓ Shall have the responsibility to present the details of all proposals for plant changes, which may affect the safety, operability and reliability of the plant/facility in any way to the Apex HSE Committee for its review and approval. All plant modifications shall be authorized as per MOC procedure.

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- ✓ Shall have the responsibility to report incidents and initiate investigation of accidents/incidents and near misses in their respective districts.
- ✓ Shall have the responsibility to handle the major emergencies arising out from their districts in accordance with the major emergency management plan.
- ✓ Shall have the responsibility to conduct periodic emergency preparedness drills as per the guidelines given in the ERP procedure.
- ✓ Shall have the responsibility to ensure that joint safety audits are carried out in their areas.

06. Ownership:

Area Owners are also the owners of various areas/facilities in their respective district with the following protocol on common facilities:

Sr No.	Common Area/Facilities	Ownership
1.	Process pipelines up to the battery limit of receiving plants	Supplier of the product
2.	Utility pipelines up to the battery limit of the receiving plants	Supplier of concerned Utility
3.	Underground sanitary sewers	
	ISBL	Individual plants
	OSBL	ADMN (get help from Area Owner)
4.	Fire water network	
	ISBL (Permit Issuer: Individual Plant and Acceptor: Engineering)	Individual Plant
	OSBL (Permit Issuer: Individual Area Owner in consultation with EHS and Acceptor: Engineering)	Individual Area Owner
5.	Fire Water Network Inspection of Various Fire Protection Equipment	HSEF dept.
	Fire Water Network Maintenance of Various Fire Protection Equipments and connected fittings.	Engineering
6.	HT Electrical cables up to the plant substations	Electrical Dept.
7.	Emergency siren testing.	HSEF Dept.
8.	Emergency siren maintenance.	Electrical
9.	Public Address System – Maintenance & Emergency Siren Maintenance.	Instrument
10.	Natural Gas Line – From Gas Station to Power Plant/Plants	CPP
11.	Scrap Yard	Stores
12.	Storm Water	
	ISBL	Individual Plant
	OSBL	Area Owner get the help from HSEF
13.	Roads	
	ISBL	Individual Plant
	OSBL	ADMIN (get help from Area Owner)

07. Annexure-A (Please refer drawing of Area Owner for better understanding):

Name of Safety Districts	Name of Owner	Street	Avenue
DPTFE, APTFE, APTFE Warehouse, SR1, SR2, Oven Area Ref. Section and 8 & 9 Reactor	Mr. Sandeep Tarafdar	1,2,8,11,12,22	3,5,6,7
ML Wash,SR-3/4, Polymer Godown, PFA, FEP & Micro Powder	Mr. Nilay Dani	7,9	1,2
Warehouse	Mr. Kamlesh Mishra	-	1,3,4
FKM, S&A, ETFEE Storage & Kilo Lab, Storage PVDF	Mr. T.K. Das	10,13,15,14	4,6

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Civils, Methanol Storage, Civils Utility	Mr. D.K. Singh	3,4,11,19,20,21,25	6,8
CA, Cl ₂ Bullet, Salt Storage, Salt Saturator, Flaker, H ₂ Filling	Mr. R.K. Gupta	01,18,21,22,23,26,27	6,7,8,10
Tank Farm, CaCl ₂ & FI, FI Utility, Lime Stone and Rejected Stone area	Mr. R.K. Gupta	05,06,25,30,37	08,13,14,15
A & H, AHF Unloading	Mr. Mahendra Kunpara	32,33,34,36,38	10,12,13,14
Power Plant, H ₂ Boiler, Coal storage, DM plant, NG skid	Mr. Mukesh Jain	27,28,29,34,35,39	9,11,12,15
Common Utility & Polymer Utility (TFE1&2, VDF,8&9th Reactor, DI Plant, FI, N ₂ Storage, Fire Water Pump House 1 & 2), Dehumidifiers	Mr. Devendra Bute	3,9,10,11,12,13, 14,15,16,22,26, 30,31,32,36,40	3,4,5,6,7,8, 10,12,15
HSEF Building, Lagoon Area	Mr. Sanjay Chowrasia	7,39	1,15
Switch Yard, All Electrical Sub-Station, & transformer area	Mr. Amit Tyagi	10,18,39,40	1,12
Main Administration Building (Security, Gates & Horticulture, New Canteen) Vastu well & Bhramasthan, all plant soak pit.	Mr. Anish Mehta	1,2,3,4,5,6,7, 17,18,23,24,29,40	1,12,15
Central Workshop, CES Building	Mr. Nilesh Doshi	7, 11, 12	3,6
Store & Raw Material Warehouse	Mr. Yojan Gajjar	-	3
Central Scrap Yard, Contractor Area, Contractor Shed, Contractor workshop, Contractor Feb. Yard	Mr. Bharat Khunt	7,17,40	15
Weigh Bridge & Logistic	Mr. Yojan Gajjar	01,17,40	1, 3,15,19
GFRC	Dr. BS Rawat	-	1
VDF & R-225 Plant, R142B Tank Farm	Mr. Abid Husain	23,29,16,18	6
PTFE, PTPTE	Mr. Sagar Patil	-	4
TFE 2/3	Mr. Sanjay Saxena	10,16,17,18,24	1,4,9
ETP (552 KLD, 1 MLD, Guard Pond, STP)	Mr. Manoj kumar Bidhar	24,27	9
ANNEX Plot	Mr. C L Prasad	-	-
Central lab, A/D/S/ PTFE lab, New polymer lab, FKM lab, TFE1 Lab	Mr. Rajesh Gupta	-	-
Pilot Plant	Mr. Suresh Babu	13	

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14. Frequently asked Question and Answer: -

1. What is the primary purpose of the Permit to Work (PTW) system?
 - a) To ensure work is completed quickly
 - b) To align with corporate standards and statutory requirements
 - c) To reduce paperwork
 - d) To increase production efficiency
2. Who is responsible for writing the permit and JSA in their safety district?
 - a) Permit Acceptor + Permit Issuer
 - b) Permit Holder+ Unit Head
 - c) Permit Issuer+ Approver
 - d) Unit Head + Approver
3. What must be confirmed by the Permit Issuer before authorizing work to proceed?
 - a) Availability of tools
 - b) Correct identification of hazards and adequacy of controls
 - c) Number of workers
 - d) Duration of the work
4. Which document must be prepared by the team before completing the PTW?
 - a) Safety Data Sheet (SDS)
 - b) Job Safety Analysis (JSA)
 - c) Risk rating
 - d) Standard Operating Procedure (SOP)
5. What is the validity period for a Cold Work Permit?
 - a) 24 hours
 - b) 7 days
 - c) 8 hours or shift end, whichever is earlier
 - d) 1 month
6. What must be done if the work conditions change during the course of an activity under permit?
 - a) Continue working as planned
 - b) Refer back to the Permit Issuer
 - c) Inform the Unit Head
 - d) Stop work immediately
7. What is required for any maintenance, installation, and repair work on electrical equipment?
 - a) Hot Work Permit
 - b) Cold Work Permit
 - c) Electrical Work Permit
 - d) Confined Space Entry Permit
8. Who is responsible for carrying out gas tests before and during jobs?
 - a) Permit Holder

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- b) Permit Acceptor
 - c) Gas Tester
 - d) Area Operator
9. What must be done before starting hot work in a hazardous area?
- a) Conduct a toolbox talk
 - b) Ensure the area is free from hydrocarbon
 - c) Conduct a gas test using a calibrated gas test meter
 - d) Inform the Unit Head
10. What is the role of a Fire Watch/Sta during hot work?
- a) To oversee any probability of fire and ensure safety for 30 minutes after job completion
 - b) To issue permits
 - c) To conduct gas tests
 - d) To approve the permit
11. What must be done if a permit needs to be extended beyond its initial validity?
- a) Issue a new permit
 - b) Revalidate and sign by the next shift engineer
 - c) Inform the Unit Head
 - d) Continue working without revalidation
12. What is the minimum depth for which an excavation permit is required?
- a) 10 cm
 - b) 30 cm
 - c) 50 cm
 - d) 1 meter
13. Who must be informed if there is an emergency in the area during permitted work?
- a) Permit Holder
 - b) Permit Issuer
 - c) Area Operator
 - d) Unit Head
14. What is the purpose of a Job Safety Analysis (JSA)?
- a) To identify potential hazards and suggest control measures
 - b) To increase production efficiency
 - c) To reduce paperwork
 - d) To train new employees
15. What must be done if a permit is lost?
- a) Issue a new permit
 - b) Use a duplicate copy from the permit issuer's copy
 - c) Inform the Unit Head
 - d) Continue working without a permit

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16. Who is responsible for ensuring that the work location has all necessary safeguards available?
 - a) Permit Holder
 - b) Permit Acceptor
 - c) Area Operator
 - d) Unit Head
17. What must be done before entering a confined space?
 - a) Conduct a toolbox talk
 - b) Ensure the area is free from hydrocarbon
 - c) Conduct a gas test and ensure the space is safe
 - d) Inform the Unit Head
18. What is the role of the Permit Acceptor?
 - a) To issue permits
 - b) To execute the work in full compliance with the permit requirements
 - c) To conduct gas tests
 - d) To approve the permit
19. What must be done if there is any abnormality found during the work?
 - a) Continue working as planned
 - b) Stop work immediately and inform to the permit issuer
 - c) Inform the Unit Head
 - d) Conduct a toolbox talk
20. What is the role of the Permit Holder?
 - a) To issue permits
 - b) To execute the work in full compliance with the permit requirements
 - c) To conduct gas tests
 - d) To approve the permit
21. What is the role of the Area Operator in the PTW system?
 - a) Issuing permits
 - b) Ensuring the work location has all necessary safeguards
 - c) Conducting gas tests
 - d) Approving permits
22. What must be done if the Permit Issuer delegates their authority during a period of absence?
 - a) Inform the Unit Head
 - b) Notify the custodian
 - c) Issue a new permit
 - d) Continue working without delegation
23. What is the purpose of a toolbox talk?
 - a) To increase production efficiency
 - b) To reduce paperwork

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- c) To ensure all team members understand the hazards and controls
 - d) To train new employees
24. Who is responsible for signing and accepting the permit?
- a) Permit Issuer
 - b) Permit Acceptor
 - c) Permit Holder
 - d) Area Operator
25. What must be done if there is a potential release of flammable gas or vapor in the vicinity of hot work?
- a) Stop working
 - b) Monitor the atmosphere using continuous gas monitors
 - c) Inform the Unit Head
 - d) Conduct a toolbox talk
26. What is the minimum oxygen concentration required for entry into a confined space?
- a) 18%
 - b) 19.5%
 - c) 20%
 - d) 21%
27. What must be done before starting any work involving the opening-up of equipment?
- a) Conduct a toolbox talk
 - b) First Follow the LOTO Procedure
 - c) Inform the Unit Head
 - d) Conduct a gas test
28. What is the role of the Authorized Toxic Gas/Oxygen Level Tester?
- a) Issuing permits
 - b) Conducting gas tests and ensuring safety
 - c) Approving permits
 - d) Executing the work
29. What must be done if a valve is found to be passing during isolation?
- a) Continue working as planned
 - b) Adopt appropriate measures such as tale blinding/additional valves
 - c) Inform the Unit Head
 - d) Conduct a toolbox talk
30. What is the purpose of the Permit Matrix?
- a) To list authorized persons for issuing, accepting, and approving permits
 - b) To increase production efficiency
 - c) To reduce paperwork
 - d) To train new employees

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31. What must be done if the work site is not safe for the work to proceed?
 - a) Continue working as planned
 - b) Stop work immediately and inform the Permit Issuer
 - c) Inform the Unit Head
 - d) Conduct a toolbox talk
32. What is the role of the Permit Approver?
 - a) Issuing permits
 - b) Ensuring hazards are evaluated and controls are provided
 - c) Conducting gas tests
 - d) Executing the work
33. What must be done if there is a change in weather conditions during permitted work?
 - a) Continue working as planned
 - b) Refer back to the Permit Issuer
 - c) Inform the Unit Head
 - d) Conduct a toolbox talk
34. What must be done before entering an excavation of depth more than 1.2 meters?
 - a) Conduct a toolbox talk
 - b) Ensure it is treated as a confined space and obtain a permit
 - c) Inform the Unit Head
 - d) Conduct a gas test
35. What must be done if the specified equipment or tools are not available during the course of an activity under permit?
 - a) Continue working as planned
 - b) Refer back to the Permit Issuer
 - c) Inform the Unit Head
 - d) Conduct a toolbox talk
36. What must be done if there is an emergency in the nearby area during permitted work?
 - a) Continue working as planned
 - b) Stop work and refer back to the Permit Issuer
 - c) Inform the Unit Head
 - d) Conduct a toolbox talk
37. What must be done if there is a change in the type and nature of work during permitted work?
 - a) Continue working as planned
 - b) Refer back to the Permit Issuer
 - c) Inform the Unit Head
 - d) Conduct a toolbox talk
38. What is required for any work involving the use of mobile X-ray/radiography equipment?
 - a) Cold Work Permit

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- b) Hot Work Permit
 - c) Electrical Work Permit
 - d) Confined Space Entry Permit
39. What must be done if there is a potential for gas generation due to the nature of work being performed inside a confined space?
- a) Conduct a toolbox talk
 - b) Carry out regular gas tests
 - c) Inform the Unit Head
 - d) Conduct a gas test
40. What must be done if there is a potential for flammable gas or vapor emission within 15 meters of the work site?
- a) Continue working as planned
 - b) Render the source safe by sealing
 - c) Inform the Unit Head
 - d) Conduct a toolbox talk
41. What must be done if there is a potential for hot material escaping during work?
- a) Allow the material to cool before draining
42. What to do if any plant wants to deviate PTW standard GFL Dahej site procedure for doing any job.
- a) Fill the deviation form given in PTW standard and take approval from Unit head.
43. Where we can find soft copy of all the standard?
- a) LINK:- <https://gflteams.sharepoint.com/sites/GFL-HSE/Shared%20Documents/Forms/AllItems.aspx>
44. When to fill the checklist in PTW. (PTW for Height work, hot work, confined space etc..)
- a) In PTW format "Word compliance" required additional checklist.
45. What is the format number of PTW?
- a) HSE/SAF/02
46. What is the format number of JSA?
- a) HSE/SAF/F/05
47. How to ensure employee has attain PTW standard training?
- a) Training data available with HR department.
48. SRP full form?
- a) Standard rules and procedure.
49. How to ensure medical checkup record
- a) Record available with HSE department.
50. Hot work permit approver during C shift.
- a) Approver II , in absence of Approver II NDO will sign after getting request mail from Approver II.

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Answer Key: -

Que.No	Answer	Que.No	Answer	Que.No	Answer	Que.No	Answer
1	b	11	b	21	b	31	b
2	a, c	12	b	22	b	32	b
3	c	13	b	23	c	33	b
4	b	14	a	24	b	34	b
5	c	15	b	25	a	35	b
6	b	16	c	26	b	36	b
7	c	17	c	27	b	37	b
8	c	18	b	28	b	38	b
9	c	19	b	29	b	39	b
10	a	20	b	30	a	40	b